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CONFINED DIFFUSION OF NANOPARTICLES USING iSCAT MICROSCOPY

by

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Layman Summary

Many products are made using very tiny pieces of material called nanoparticles. These are found in things like plastics, medicines, and food. To make sure these products are safe and good quality, we need to measure what these tiny pieces are like. For example, their size and how they behave.

However, nanoparticles are so small that we cannot see them clearly, even with a microscope that uses light. So we need to find new ways to study them.

This thesis develops an alternative method to do this. It measures where a particle is located in space, and it watches how that position changes over time. Using mathematics, we can then learn more about the particle from the way it moves. This helps us measure nanoparticles more accurately.

Abstract

The understanding of thermal movement of paricles within a confined space has been studied extensively in order to understand biological, chemical, and physical processes. Although the theoretical understanding of this confined diffusion is well defined in literature, not much work has been done to understnad the effects of electrostatic and hydrodynamic contributions to the size measurements measured from the motion of nanoparticles, as measurements are often instrinsically performed under confinement. This is due to the fact that 3D particle-tracking is not as commonly used as other, higher throughput sizing techniques. By employing developed techniques of nanoparticle tracking in tandem with a full theoretical model, the non-trivial effects of confinement on size-diffusivity were directly investigated and characterized. The work described in the thesis advances the method and tools of 3D nanoparticle analysis both theoretically and in application.

The first section of the thesis describes a theoretical model of Brownian motion based on the confined-Langevin model using a Debye-Huckel electrostatic potential. The model was implemented, based on literature [1], into a computer-based Python simulation in order to parameterize and create prior distributions of particle motion.

The second section describes a modified method of 3D tracking to obtain faster calculations of particle position from optical microscopy images. This was based on an algorithm described in literature [2]. The validation of the method was done on exact point-spread function simulation data with an accuracy of 1 nm.

The third section describes the aquisition of experimental data using a custom interferometric scattering microscope (iSCAT) with image pixel resolution 55 nm aquiring images between 400 and 1000 frames per second. The data consisted of images of scattering fringes from nanoparticle species and was analyzed using the algorithm developed in section 2. The resultant information was accuracte 3D particle tracks. The models developed in sectino 1 were applied to the tracks to obtain infer particle size and charge.

The thesis presents a method of finding particle size, motion, and electrostatics which can present additional tools to analyze nanoparticle characterization, which was the goal of the thesis.

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Nomenclature

Symbol	Description
δ_{ij}	Kronecker delta
$\delta(\cdot)$	Dirac delta distribution
$\vec{x}$	Position vector
$\vec{v}$	Velocity vector
m	Mass
t	Time
κ	Inverse Debye length
ζ_p	Particle zeta potential
ψ_w	Wall zeta potential
Γ_i	Stern layer of species i
Γ	Fluctuation–dissipation magnitude
j	Probability current
W_t	Gaussian white-noise process
σ_w	Surface charge density
D	Diffusion coefficient
h	Length
Ψ	Electrostatic potential
ρ	Volumetric charge density
ε_0	Permittivity of free space
ε_r	Relative permittivity of the material
Z	Ion number (valence)
C	Concentration

Symbol	**Description**
β	$1/k_bT$
e_0	Elementary charge
a	Spherical particle radius
η	Fluid viscosity
k_b	Boltzmann constant
T	Temperature
$\mu_\parallel$	Parallel (axial) mobility coefficient
$\mu_\perp$	Perpendicular (transverse) mobility coefficient
Re	Reynolds number

Chapter 1

Introduction

Objects in nature fall into two size-categories: the macroscopic and the microscopic. Macroscopic objects can be directly measured with a ruler or a photograph as they are much larger than compared to light. However, as the size drops below the wavelength of visible light, $\sim 200\,\text{nm}$, they become un-resolvable by the light alone. Due to the Abbe diffraction limit, the smallest resolvable size of a particle is half of the wavelength of light [3, 4]. Recovering the dimensions of anything smaller, or sub-diffraction, requires indirect measurements.

The Abbe diffraction limit plays a significant role in observing biological systems with sizes that are sub-diffraction. In biological systems, size of proteins, enzymes, and organelles are not only a geometric property, but they are also relate to the functional properties. Thus, the size is a critical parameter for biological systems.

For mRNA-lipid nanoparticles (mRNA-LNP), whose therapeutic effect depends on how efficiently the mRNA is released into the cellular environment, size is utmost importance. The release of mRNA into the cytoplasm depends on a size-selective biological process known as endosomal escape [5–9]. Since the potency of the mRNA-LNP varies with a parameter related to the size of the particle, observing it in the biological environment gives incredible insight to the chemical process. Thus, the precise measurement of particle size is essential to reliable pharmacological production, quality control, and rational design.

One universal property of sub-micron particles is that they undergo an apparent size-related stochastic motion when suspended in media [10–12]. At the nano-scale, thermal energy, k_bT, is the dominant force that acts on the particle. The gravitational and inertial energy, that typical govern macroscopic dynamics, are negligible in this length-scale. In liquid, or continuous media, the viscous drag dominates over the inertial force, which constitutes a memory-less, random motion [1]. The emergence of this thermal motion occurs when the ratio of the inertial forces to

[1] This is also a function of temporal resolution [12].

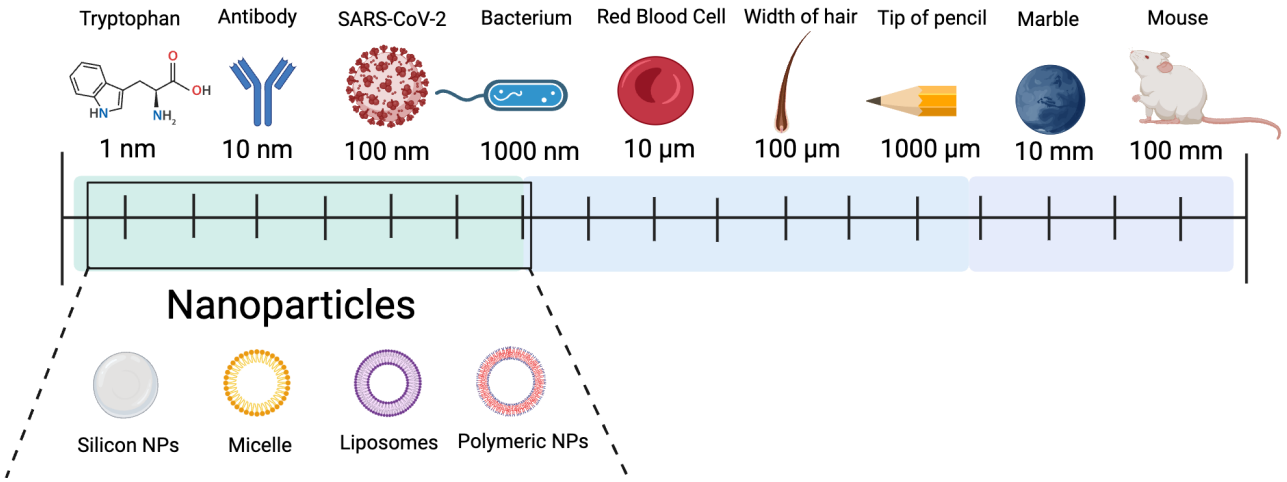


Figure 1.1: Size comparison of various particles. Nanoparticles are defined as particles where the largest dimension is at most $\sim 200\,\text{nm}$ REF.

viscous forces, known as the Reynold's number, is much lower than one [12],

$$\text{Re} \ll 1.$$

The relation between the particle's size and its motion is explained in Einstein's seminal paper [11]. The paper states the rate of transport, or diffusion, is related to the particle's hydrodynamic radius,

$$D(\vec{x}, a) = \frac{k_b T}{\gamma(\vec{x}, a)}, \tag{1.1}$$

where $\gamma(\vec{x}, a)$ is the viscous drag at the position $\vec{x}$ of a particle of hydrodynamic radius a. In the case of a spherical paricle in an incompressible fluid, $\gamma(a) = 6\pi\eta a$, where η is the fluid viscosity.

In addition to the diffusion relation, Einstein's paper also described the relationship between the mean-squared displacement and the diffusion coefficient,

$$\langle x^2 \rangle_c = 2dD(\vec{x}, a)\Delta t, \tag{1.2}$$

where $\langle x^2 \rangle_c$ is a statistical property of a particle motion, and Δt is time between steps [11, 12]. Thus, Einstein showed that the diffusion coefficient, and implicitly the size, can be extracted from the variance of particle displacement.

In order for the size estimate obtained from stochastic particle trajectories to be statistically relevant, the number of data points in the positional trajectory must be sufficient. Since the statistical uncertainty of the diffusion coefficient scales as $\sim \sqrt{N}$, a higher number of particle locations yield a more precise size inference [13].

Long particle trajectories are typically obtained by imaging the particle position over a period of

time using optical microscopy. One method of optical microscopy is to use fluorescence microscopy. Fluorescence microscopy utilizes covalently bonded fluorophore dyes to produce optical signal under laser excitation. Although fluorescence signal is high, the fluorophores have finite lifetimes, to which they oxidize and stop producing light, known as photobleaching [14]. Alternatively, scattering microscopy is a technique that utilizes the scattering of light from particle change in refractive index to obtain signal. This technique requires computational processing to realize signal. This process can be imaged indefinitely without loss in signal as particles are not damaged by the light alone.

The requirement of having many observations favors the use of a scattering microscopy over fluorescence microscopy. In addition, fluorescent microscopes, although possible to obtain three-dimensional position, are often only accurate for two-dimensional localization REF. In addition, many commonly used optical methods are limited to only two-dimensional slices of a focal plane, which is typically on the order of a few hundred nanometers.

In order to obtain large number of particle positions, while maintaining independent particle specificity, particles are often spatially confined within the field of view of an optical imaging system for an extended period of time REF?. Without axial confinement, particles rapidly diffuse out of the focal plane. In recent scientific publications, convex-lens induced confinement (CLiC) measurements have typically utilized micro-geometries of volume $\sim 1.5\,\mu m^3$ in order to confine single particles for temporally extended measurements via fluorescence microscopy [15–17]. In complement, Nanoparticle tracking analysis (NTA) measurements follow the same principles, where flow is applied to particles confined between two plates and are imaged using side-scattering microscopy [18]. In NTA, however, particles are still able to diffuse in and out of the focal volume.

Although experimentally feasible, confinement in small spaces influences the motion of particles near surfaces. This property intrinsically modifies the size-diffusivity relation. This effect is non-trivial, introducing a position dependent viscous drag that has been shown to alter the diffusivity by a significant amount [12, 19].

To account for near-wall effects, the particle must be tracked in three-dimensions with respect to the confinement barriers. Two dimensional tracking, which is commonly performed with optical microscopy alone, is theoretically insufficient to recover the true size and to find spatially varying diffusion coefficient REF. Without the axial dimensions, the spatially dependent correction cannot be inferred.

To measure the nanoparticle properties, such as diffusivity and size, this thesis presents an advancement of the iSCAT technique allowing for rapid and accurate three-dimensional tracking reconstruction. From the trajectory, the diffusivity, size, and environment conditions can be easily determined.

Chapter 2

Brownian Motion

The physical phenomenon of irregular movements of nanoparticles in liquid is known as Brownian motion. The observation of this motion originated around the year 1827 when Robert Brown observed a stochastic motion of clay particles originating from plant pollen under his microscope [10]. This stochastic motion is known as *Brownian Motion* and is invertibly related to the size of the particle.

For relatively large particles, the optical microscope is a good tool to observe them. However, for particles smaller the nano-scale, the diffraction limit is a property that must be overcome. This was theoretically described by Abbe [3, 4].

This chapter aims to establish the theoretical background underlying the relationships between the diffusivity, particle size, and spatial coordinate. This process requires sufficient mathematical detail as no single displacement carries enough information to define the whole trace, however, as one observes this motion for a longer period of time, the randomness tends to converge to a lawful, physical origin.

This chapter introduces the mathematics of Brownian motion and diffusion-type processes. Einstein's diffusion- equation argument and Langevin's stochastic equation of motion are discussed and shown to be equivalent through the Fokker-Planck formalism. In the case of diffusion under confinement, where hydrodynamic wall effects modify the mobility and electrostatic interactions, described by the Debye-Hückel interactions, govern particle-wall interactions in ionic solution. Finally, the subject of equilibrium distributions are discussed with regards to spatial probability distributions arising in confined diffusion processes.

2.1 Classical Description of Einstein's Brownian Motion

The physical phenomenon of diffusion can be rigorously described as the emergent behavior atomic particle undergoing random collisions with members of the surrounding environment. In

the microscopic limit, where the radius of a particle is sufficiently small and thermal fluctuations dominate its dynamics, this erratic motion is referred to as Brownian motion, or more colloquially, a diffusive process [10].

The classical description of Brownian motion is concerned with how the molecular-kinetic theory of heat relates with microscopic particle suspensions in liquid or gas. In particular, how do these particles perform movements such that they can be viewed by a microscope? Einstein's argument is one of macroscopic motion. Mathematically, one can use a Taylor series expansion over a general concentration function to show a natural property of the system. Using Einstein's original notation, suppose that $\phi(\Delta)$ denotes the probability to find a particle in some space Δ subject to the normalization

$$\int \phi(\Delta) d\Delta = 1 \tag{2.1}$$

Assume that the medium is isotropic such that ϕ is even, $\phi(\Delta) = \phi(-\Delta)$.

Let $f(x,t)$ denote some number of particles at the point x. Einstein takes a finite slice of this volume and equates it to a probability space;

$$f(x, t+\tau) = \int f(x+\Delta, t)\phi(\Delta) d\Delta \tag{2.2}$$

The argument here is that we can taylor expand both sides of this equation,

$$f(x, t+\tau) \approx f(x,t) + \tau \frac{\partial f(x,t)}{\partial t} \tag{2.3}$$

And in space;

$$f(x+\Delta, t) \approx f(x,t) + \Delta \frac{\partial f(x,t)}{\partial x} + \frac{\Delta^2}{2} \frac{\partial^2 f(x,t)}{\partial x^2} \tag{2.4}$$

Cancelling terms, one finds the classical diffusion equation to nearest order;

$$\frac{\partial f(x,t)}{\partial t} = \underbrace{\left[\frac{1}{\tau} \int \frac{\Delta^2}{2} \phi(\Delta) d\Delta \right]}_{D} \frac{\partial^2 f(x,t)}{\partial x^2} \tag{2.5}$$

Where we typically associate the diffusion coefficient with the second moment of the probability distribution

$$D = \frac{1}{\tau} \int \frac{\Delta^2}{2} \phi(\Delta) d\Delta \tag{2.6}$$

The friction coefficient here is related to the solution of the Navier-Stokes equations at a low Reynolds number. Specifically, when there are no inertial forces and $\mathrm{Re} \ll 1$, then the drag force

on a particle,

$$\vec{F}_d = 6\pi\eta a v, \tag{2.7}$$

where $\gamma_0 = 6\pi\eta a$, η is the fluid viscosity, and a is the spherical particle's radius. This relation can be directly applied to the form of the diffusion coefficient to recover Einstein's famous formula;

$$D = \frac{k_b T}{6\pi\eta a}. \tag{2.8}$$

2.2 Langevin Dynamics

In typical experiments, such as Robert Brown's particulates or diffusing lipid-nanoparticles, the observed Brownian particle move within a fluid volume. If one assumes that this particle exists in the thermodynamic equilibrium, it should obey the equipartion theorem. However, the observed motion appears to have more energy than predicted by the equipartion at certain times, indicating that an additional contribution to the dynamics must be considered REF.

This missing component of motion can be described by the bombardment of the surrounding particles in the liquid. In particular, the system experiences a *random* force on it at every time step.

$$m\ddot{\vec{x}} = -\frac{\dot{\vec{x}}}{B} + \vec{F}(t) \tag{2.9}$$

where B is the mobility (drag term is $-\frac{\dot{\vec{x}}}{B}$), which is represented in a viscous friction term, and $\vec{F}(t)$ is a stochastic force term, meaning it takes a random value from some distribution f for every time step t. This model, although easily written, is quite profound and its analytical treatment is non-trivial. Here, $x(t)$ is a stochastic process, where each realization of $F(t)$ produces a unique trajectory of $x(t)$.

The statistical content of $\vec{F}(t)$ is determined by the moments and auto-correlation. Notably, this relation is physically described by the fluctuation-dissipation theory, which relates the drag force to the random force. In Brownian motion, this takes the form

$$\langle \vec{F}(t) \rangle = 0, \quad \langle \vec{F}_i(t) \vec{F}_j(t') \rangle = \Gamma \delta_{ij} \delta(t - t') \tag{2.10}$$

Which stands in as delta-correlated noise, or that of a Wiener process. Notably, Γ is constrained by the equipartion theorem. This can be shown by calculating the velocity autocorrelation function (VACF) and choosing the large t limit to agree with the equipartion theorem in 3-dimensions. This fixes the fluctuation constant to being

$$\Gamma = \frac{2k_b T}{B}. \tag{2.11}$$

One can take the ensemble average of this device, and notably, extract correlation information. Rewriting in terms of velocity, and assuming zero drift velocity one finds

$$m\dot{\vec{v}} = -\frac{\vec{v}}{B} + \vec{A}(t), \quad \int_{-\infty}^{\infty} A(t)dt = 0. \tag{2.12}$$

This represents a linear differential equation with a solution hat defines the velocity autocorrelation function. This solution is

$$\langle \vec{v}(0) \cdot \vec{v}(t) \rangle = \frac{3k_bT}{m} e^{-t/(mB)}. \tag{2.13}$$

Furthermore, when one averages the product of the above equation with $\vec{x}$ and integrates over time, we find the following form of the mean-square displacement. Note that $\langle v^2 \rangle = \frac{3k_bT}{m}$, then the solutions of the position mean-squared displacement are written as

$$\begin{aligned} \langle x^2 \rangle &= 6k_bTmB^2 \left[\frac{t}{mB} - (1 - e^{-t/(mB)}) \right] \\ &\begin{cases} \xrightarrow{t \ll mB} \frac{3k_bTt^2}{m} & = \langle v^2 \rangle t^2 \\ \xrightarrow{t \gg mB} \frac{6k_bTmB}{m} t & = (6Bk_bT)t \end{cases}. \end{aligned} \tag{2.14}$$

Letting $D = Bk_bT$, we recover Einstein's relation with ease.

The unique advantage of the Langevin Formulation is that it is readily simulate-able [1]. It can be shown that using an Euler-Maruyama scheme, the stochastic differential equation can be integrated over realizations of $F(t)$ using a random number generator, recovering the dynamics of an ideal brownian system [20].

2.3 From Master Equation to Fokker-Planck

To show that the two methods are equal, we can use the typical approach to equilibrium provided by the master equation. Suppose that $f(x,t)dx$ representation that he probability of an arbitrary particle in the ensemble has a displacement between x and $x + dx$. We find that if f is a probability distribution, we can write down the master equation, let W represent the traditional probability transition rate;

$$\frac{\partial f(x,t)}{\partial t} = \int_{-\infty}^{\infty} \left[f(x',t)W(x'|x) - f(x,t)W(x|x') \right] dx' \tag{2.15}$$

This structure is a statement about the probability of positional transitions, which is fundamentally equivalent to the Langevin equation[1] The right hand side can be expanded using the Kramer-Moyal

[1] Given a Markovian system.

expansion and truncated at second order, yielding the Fokker-Planck Equation.,

$$\frac{\partial f(x,t)}{\partial t} = -\frac{\partial}{\partial x}\left[\sigma(x) f(x,t)\right] + \frac{1}{2}\frac{\partial^2}{\partial x^2}\left[D(x) f(x,t)\right], \tag{2.16}$$

where the diffusive and drift terms are parameterized over the transition probabilities,

$$\begin{aligned} \sigma(x) &= \int_{-\infty}^{\infty} \xi W(x|\xi) d\xi \\ D(x) &= \int_{-\infty}^{\infty} \xi^2 W(x|\xi) d\xi. \end{aligned} \tag{2.17}$$

In the case of zero drift, we can clearly see that this form equates to the diffusion equation of the Langevin formulation and the Einstein-ian formulation are the same. This important property allows us to investigate solutions and dynamics of the two systems interchangeably. In particular, relating expected results from simulation-based behavior of Langevin problems.

2.4 Confined Diffusion Processes

The effect of diffusion near an interface on a nearby particle is quite particular. The effect on confinement can be first imagined as the effects from the fluid friction flow. Notably, the particle being immersed in fluid invites a stokes friction which effects the perpendicular and parallel flows. Hiding Faxén derived the relations in no-slip boundary conditions near a brownian particle;

$$\begin{aligned} \mu_{\parallel}\left(\frac{a}{h}\right) &\approx \mu_{\infty}\left[1 - \frac{9}{16}\left(\frac{a}{h}\right) + \frac{1}{8}\left(\frac{a}{h}\right)^3 - \frac{45}{256}\left(\frac{a}{h}\right)^4 - \frac{1}{16}\left(\frac{a}{h}\right)^5\right] \\ \mu_{\perp}\left(\frac{a}{h}\right) &\approx \mu_{\infty}\left[1 - \frac{9}{8}\left(\frac{a}{h}\right) + \frac{1}{2}\left(\frac{a}{h}\right)^3\right] \end{aligned} \tag{2.18}$$

In the case of a two-wall confinement, the form of the diffusion coefficient has a lot of depth in wealth. In application, it has been shown that the form of the Onseen drag in terms of mobility coefficients.

$$\gamma_2(z) \approx \gamma_0 + [\gamma_1(z) - \gamma_0] + [\gamma_1(H - z) - \gamma_0] \tag{2.19}$$

2.5 Electrostatic Potentials

A particle diffusing near a charged wall in an electrolyte will feel a screened electrostatic interaction. The Debye-Huckel theory is a description of the electromagnetic interactions between ions in a solution of ionic compounds. The potential in a field of ions must be solved in terms of

electrostatics in a media. This is described by Poisson's equation

$$\nabla^2\Psi(\vec{x}) = -\frac{\rho(\vec{x})}{\varepsilon_0\varepsilon_r} \tag{2.20}$$

In a sea of ions, we must carefully define the charge density as it pertains to the system. In a general solution of ions, there will be j species of ions.

$$\nabla^2\Psi(\vec{x}) = -4\pi\frac{\rho_j(\vec{x})}{\varepsilon}$$

The charge distribution of a single ion concentration is essentially the density of the ionic positions. It would be tedious and unfeasible to count every ion's position exactly as we are assuming too many things. Rather, one can use a statistical ensemble of ions via a canonical ensemble, which is a good approximation here.

$$\rho_i(\vec{x}) = \sum_i C_i e_0 Z_i \exp\left[-\beta Z_i e_0 \Psi(\vec{x})\right] \tag{2.21}$$

Here the energy of the system is defined as the potential energy scaled by the charge defined as $U = qV$. The full equation is known as the Poisson-Boltzmann equation, where $\Psi(\mathbf{r})$ is the potential for all space.

$$\nabla^2\Psi(\vec{x}) = -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} C_i e_0 Z_i \exp\left[-\beta Z_i e_0 \Psi(\vec{x})\right] \tag{2.22}$$

There is a nice approximation we can make given that the potential of the solution is small. We linearize the exponential to first order here.

$$\begin{aligned}
\nabla^2\Psi(\vec{x}) &\approx -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} C_i e_0 Z_i \left[1 - \beta Z_i e_0 \Psi(\vec{x})\right] \\
&= -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} \left[C_i e_0 Z_i - \beta C_i Z_i^2 e_0^2 \Psi(\vec{x})\right] \\
&= -\sum_{i=1}^{\infty}\left[\frac{C_i e_0 Z_i}{\varepsilon_0\varepsilon_r} - \kappa_i^2\Psi(\vec{x})\right]
\end{aligned}$$

Where κ_i^2 is related to the debye-length, a unit of the electrostatic screening caused by the ionic interactions. The Debye length is defined as $\lambda_D = \kappa^{-1}$.

When the bulk solution is electro-neutral, $\sum_i C_i Z_i = 0$, then the zeroth therm can cleanly drop

out, leaving only the approximation;

$$\nabla^2\Psi(\vec{x}) = \kappa^2\Psi(\vec{x}), \quad \kappa_i^2 = \frac{\beta e_0^2}{\varepsilon_0\varepsilon_r}\sum_i C_i Z_i^2. \tag{2.23}$$

This equation has an obvious solution which are the real exponential functions;

$$\Psi(\vec{x}) = Ae^{-\kappa\vec{x}} + Be^{\kappa\vec{x}}. \tag{2.24}$$

The pre-factor A, B is the strength of the electric potential. In the case of ionic potentials, the strength of the pre-factor is a number of the number of ions and their charge density surrounding the surface. Thus, one can parameterize this pre-factor based on the geometry.

In the approximations done in this thesis, we assume a linear symmetric potential, as well as a spherical particle between them. The form of this potential is given by

$$B = C\Gamma_p\Gamma_w, \tag{2.25}$$

where $|\psi|$ is a fixed amplitude due to charge density, and Γ_w, Γ_p are terms associated with the Stern layers of the wall and the particle;

$$\begin{aligned} C &= \frac{64\pi\varepsilon_0\varepsilon_r a k_b T}{e_0^2} \\ \Gamma_w &= \tanh\left(\frac{e\psi_w}{4k_bT}\right) \\ \Gamma_p &= \tanh\left(\frac{e\zeta_p}{4k_bT}\right) \\ \psi_w &= \frac{2k_bT}{e}\text{asinh}\frac{e\lambda_D\sigma_w}{2\varepsilon_0\varepsilon_r k_bT}. \end{aligned} \tag{2.26}$$

Note that ψ_w is the Stern layer of the wall related to the surface charge density of the wall [21]. This is chosen for a more intuitive measurable of a wall-like structure.

2.6 Equilibrium Distributions for Confined Systems

In the case of interfaces, we have boundary conditions that case the particle to reflect off of the sides. Notably, the probability flux is set to 0. This implies that $\partial_x j = 0$ as well. Then, we need to solve the adjusted Fokker-Plank equation;

$$j = -\frac{\partial}{\partial x}\left[D(x)f(x)\right] - \nabla V(x)f(x) \tag{2.27}$$

Setting j to 0, we find that the solution of this equation defines the equilibrium distribution after

a long time.

$$f(x) = \frac{\exp\left[-\beta \int_0^x (\frac{\nabla V(y)}{D(y)})dy\right]}{\exp\left[-\beta \int_{-H}^{H} (\frac{\nabla V(y)}{D(y)})dy\right]} \tag{2.28}$$

This equation defines the equilibrium distribution of the brownian diffusion after a long time.

Chapter 3

Biological Application

As we have described, the diffusivity of a spherical particle is directly linked to a motion that is thermally driven. It happens that in the nature of biological cells, this diffusive motion is a driver for many reactions, such as enzyme-protein coupling, protein-protein coupling, and vesicle binding. Thus, it is of utmost importance to adequately understand the confinement effect of diffusion in a given system.

Application-wise, the confinement of nanoparticles is directly related to fabrication and development research. Notably, since nanoparticle fabrication involves methods of quality control, these methods must be able to quantify between different measures of noise impending their measurement. In the case of confinement effects, these effects often hinder diffusion, as described, which can indirectly bias the final reported measurement.

3.1 Diffusion in cells

The cellular system consists of many different components co-existing in the cellular cytoplasm. To get a survey of sizes, the organelles are on the order of $1 - 10\,\mu\text{m}$. while smaller proteins and enzymes, such as ribosomes, vesicles, and lysosomes are on the order of $10 - 100\,\text{nm}$.

This beings said, the motion is directly linked to the fluidic properties of the cytoplasm. In the case of a eukaryotic cell, the Reynold's number is low, which implies an overdamped langevin equation. In addition, the thermal scale is that of high temperature, or $\sim 310\,\text{K}$. In addition, the comparison between diffusion and advection is important. For diffusion to dominate we need that the diffusive timescale is must shorter. Here, we find that since the Peclet number is proportional to the characteristics length L, while the reynolds number is L^2. for small characteristic lengths, such as those proposed, we see that the diffusive term is clearly shorter.

3.2 Brownian motion as a method for measurements of size in biology

As previously shown, we can infer the stoke's einstein diffusion relation to directly measure the size of a biological diffusing particle. In particular, this relation can be explicitly formalized for the hydrodynamic radius of a spherical nanoparticle.

$$a = \frac{k_b T}{6\pi\eta D(\vec{x})} \tag{3.1}$$

Methods such as nanoparticle-tracking analysis (NTA) or single-molecule particle tracking (SPT) are based on this method.

However, there are other methods that can be used. As this thesis is not concerned with fluorescent methods, they will not be explained in depth, however, fluorescence correlation spectroscopy (FCS) works on similar ideas of measuring diffusion autocorrelation curves using a fluorescent probe.

In compliment to single molecule, there are bulk methods as well. In particular, dynamic light scattering (DLS) allows to measure the size of molecules based on their shape function. In particular, the scattering intensity can be weighting to the inverse of a^6, which in turn returns the size of the sample. However, these methods are ill-posed for poly-disperse sample, which is one of the weaknesses of the DLS method.

3.3 Lipid Nanoparticles

The biological molecule of choice for this investigation of confinement is a useful one with lots of applications - the lipid nanoparticle (LNP).

First, it is necessary to identify with the liposome. The liposome, as a macroscopic object, is a spherical membrane of lipid subunit, each carrying a charge and hydrophobicity. The size of a single submit is on the level of angstroms, however, the final liposome can be between $1\,\text{nm} - 10\,\mu\text{m}$. Liposomes and cellular vesicles are very similar, which naturally posed the question of drug delivery using a liposome model.

In recent years, the LNP was formulated - a collection of lipid based walls that are used to encapsulate a drug or cargo. These have applications in drug delivery and have been using in the Sars-Covid vaccine in 2020. These molecules are very useful and thus it is important to know how to improve them.

Modern tools to image and characterize the LNP are split into single-molecule techniques and bulk assays. In the realm of single-molecule methods, CLiC has been shown to be a high-throughput, accurate, and reliable method to characterize these particle's properties.

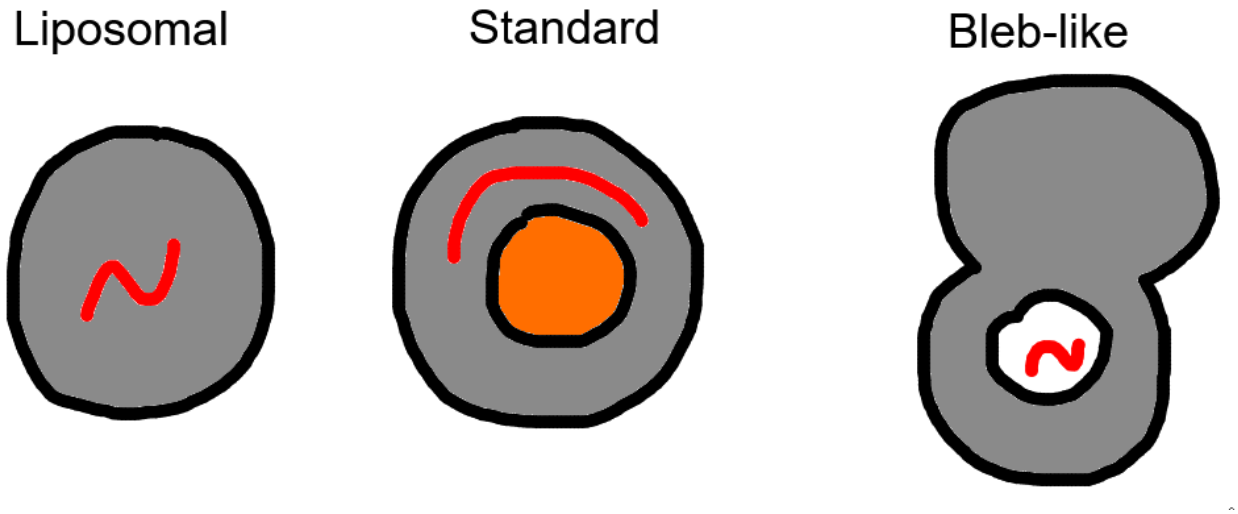


Figure 3.1: mRNA-LNP types diagram. Black is lipids, grey is hydrophobic region, red is mRNA, orange is oil-core.

3.4 LNP topology types

One recent investigation of LNP research is within the functional differences based on the LNP topology type. These topologies have many different types, but typically the topologies are grouped into three main types - liposomal, spherical, and bleb-like. These three topologies not only have different structures, but also different length scales. The topologies are visualized in the figure below.

Chapter 4

Interferometric Scattering Principles

To precisely determine the size of nanoscale particles,it is essential to obtain long and procise positional trajectories. To achieve this, a particle localization method that maintains a consistent signal over extended measurement periods is required. One method that enables such localization of particle traces is interferometric scattering microscopy (iSCAT) [14].

iSCAT is a relatively modern optical imaging technique that utilizes light scattering contrast to localize particle signal. The particle's transverse position can be directly rendered by utilizing the scattering pattern's angular symmetry. Similarly, utilizing the phase difference in the scattering pattern between series of images the relative axial distance can be measured. Thus, the 3D positional information is encoded in the scattering pattern in iSCAT.

The phase information stored in iSCAT images contain relative axial position which is more consistent and reliable relative to other techniques REF.

We begin with the basics of relating light to electromagnetism, starting with the Maxwell and Helmholtz relations. We continue discussing the scattering of a various size range of spheres, which is the approximation we take in this thesis. We then discuss the various aberrations and Considerations before introducing iSCAT microscopy. We tease the experimental apparatus used in this work and compare the method to other methods currently used in the field. Finally, the electromagnetic theory of scattering off spheres, discussion of general optical scattering microscopy theory, and principle of iSCAT will be discussed.

4.1 Light as a casual field

In order to discuss the spatial distribution of light scattering, it is necessary to use a framework to develop the fundamental properties. On the quantum level, electromagnetic wave field consist of photons [22]. This fact motivates the use of the theory of electromagnetic waves, in particular Maxwell's equations, to be applied to optical field. The optical field is defined with respect to

light intensity, which is defined as the modulus of itself,

$$I = |\vec{E}(\vec{x}, t)|^2, \tag{4.1}$$

where $\vec{E}(\vec{x}, t)$ is defined as the optical field[1]. This principle relation allows us to form models using electromagnetic theory to map out how the optical fields behave.

4.2 Electromagnetic Scattering of of Homogenous Spheres

In iSCAT, the light collected at the camera is originated from light scattering off of sphere-like particles and mapped directly to axial position. To simplify the mathematics and obtain positional information from such scattering fields, the scattering dynamics can be modelled as spheres.

The optical theory of light will obey the Maxwell's equations, which motivates solutions to obey the wave equation for the electric and magnetic Fields

$$\nabla^2 \vec{E} + k^2 \vec{E} = 0, \quad \vec{H} = \frac{1}{i\omega\mu_0} \nabla \times \vec{E}. \tag{4.2}$$

The eigenfunctions of the wave equation are the harmonics of the Laplacian operator, ∇^2, in an arbitrary coordinate basis. For isotropic scattering, radial scattering, the spherical basis has been used in literature [24]. The solutions of the vector-wave equation are known as the vector spherical harmonics.

Let ψ be a solution to the scalar wave equation,

$$\nabla^2 \psi + k^2 \psi = 0, \tag{4.3}$$

then the generator of the general solution is taken by sum of components. This the solution is given by the separable form[25],

$$\begin{aligned} \psi(r, \theta, \phi) &= \Phi(\phi)\Theta(\theta)R(r), \\ \Phi(\phi) &\in \{\cos(m\phi), \sin(m\phi)\}, \\ \Theta(\theta) &= P_n^m(\cos\theta), \\ R(r) &\in \{j_n, y_n, j_n + iy_n, j_n - iy_n\}. \end{aligned} \tag{4.4}$$

Here, the symbols j_n, y_n represent the spherical Bessel functions. Their complex sums, $j_n \pm iy_n$, are the hankel functions. The vector spherical harmonics are best described by separating these solutions into even and odd solutions as their pre-factor will be cos or sin respectively, found in

[1] As opposed to an electric field, where $\vec{E}_{\text{optic}}(\vec{x}, t) = \sqrt{\varepsilon_0 n_m c/2}\vec{E}_{\text{elec}}(\vec{x}, t)$. The normalization constant affects how the fields relate to the measured light intensity [23]

Eqns 4.17 to 4.20 in Bohren [24].

$$\begin{aligned} \vec{M}_{emn} &= \vec{\nabla} \times (\vec{r}\psi_{emn}), \quad \vec{M}_{omn} = \vec{\nabla} \times (\vec{r}\psi_{omn}), \\ \vec{N}_{emn} &= \frac{1}{k}\vec{\nabla} \times \vec{M}_{emn}, \quad \vec{N}_{omn} = \frac{1}{k}\vec{\nabla} \times \vec{M}_{omn.} \end{aligned} \tag{4.5}$$

A scattering solution will consist of incident and scattering light. Here, Tte incident light can be assumed to be a polarized plane wave along the x axis without loss of generality. The general form of these equation are then the sum of the spherical vector harmonics. When subject to the energy conservation boundary condition,

$$(\vec{E}_i + \vec{E}_s - \vec{E}_1) \times \hat{r} = (\vec{H}_i + \vec{H}_s - \vec{H}_1) \times \hat{r} = 0, \tag{4.6}$$

Utilizing the orthogonality of the spherical vector harmonics, one arrives are the exact solution known as the Mie Theory form of scattering off of a sphere [24],

$$\begin{aligned} \vec{E}_i &= E_0 e^{ikr\cos\theta}\hat{x} \\ \vec{E}_s &= \sum_{n=1}^{\infty} E_n(ia_n \vec{N^{(1)}}_{e1n} - b_n \vec{M^{(1)}}_{o1n}), \\ \vec{E}_1 &= \sum_{n=1}^{\infty} E_n(ic_n \vec{M^{(1)}}_{o1n} - d_n \vec{N^{(1)}}_{e1n}). \end{aligned} \tag{4.7}$$

In this thesis, only the scattering field is of importance. The amplitude factors of interest are,

$$\begin{aligned} a_n &= \frac{m\Psi(mx)\Psi'(x) - \Psi(x)\Psi'(mx)}{m\Psi(mx)\xi'(x) - \xi(x)\Psi'(mx)}, \\ b_n &= \frac{\Psi(mx)\Psi'(x) - m\Psi(x)\Psi'(mx)}{\Psi(mx)\xi'(x) - m\xi(x)\Psi'(mx)}, \end{aligned} \tag{4.8}$$

where Ψ, ξ are the Riccatti-Bessel functions [24].

4.3 Rayleigh Scattering

In the limit of a small particle from far away, we can mae some cancellations in these constants to simplify the work. Notably, the Hankel function approaches the limit of a spherical wave,

$$h_n(kr) \xrightarrow{kr \gg 1} (-i)^{n+1}\frac{e^{ikr}}{kr} \tag{4.9}$$

In this length-scale, we have that $a_1 \approx \frac{-2i}{3}(ka)^3\frac{m^2-1}{m^2+2}$, and $b_1 \approx 0$. Thus for $n = 1$ we have that

$\pi_1 = 1, \tau_1 = \cos\theta$, thus

$$\begin{aligned} M^{(3)} &\approx (-i)^{n+1}\frac{e^{ikr}}{kr}\left(\cos\phi\pi_n\hat{\theta} - \sin\phi\tau_n\hat{\phi}\right) \\ N^{(3)} &\approx (-i)^{n+1}\frac{e^{ikr}}{kr}\left(\cos\phi\pi_n\hat{\theta} - \sin\phi\tau_n\hat{\phi}\right) \end{aligned} \tag{4.10}$$

can be written in the Rayleigh approximation in the form of the full scattering field,

$$\vec{E}_s = \frac{iE_0e^{ikr}}{kr}\left[\hat{\theta}\cos\phi\sum_{n=1}^{\infty}\frac{2n+1}{n(n+1)}(a_n\tau_n + b_n\pi_n) + \hat{\phi}\sin\phi\sum_{n=1}^{\infty}\frac{2n+1}{n(n+1)}(a_n\pi_n + b_n\tau_n)\right]. \tag{4.11}$$

After subtituting the terms for $n = 1$, there is a neat simplification,

$$\begin{aligned} \vec{E}_{s,\perp} &= \vec{E}_{i,\perp}\frac{k^2\alpha e^{ikr}}{r}, \\ \vec{E}_{s,\parallel} &= \vec{E}_{i,\parallel}\frac{k^2\alpha e^{ikr}}{r}\cos\theta. \end{aligned} \tag{4.12}$$

Singling out the scattering field and noting the form of the wave-vector, $k = \frac{2\pi}{\lambda/n}$, one finds the final scattering field of spheres on the order of the length of the wavelength of the incident light,

$$\vec{E}_s = \vec{E}_0e^{ikr}\frac{\sqrt{8}\alpha\pi^2n_m^2}{\lambda^2r}\sqrt{1+\cos\theta}. \tag{4.13}$$

4.4 Scattering Microscopy

As shown, the description of the scattered light from a sphere is full described by Eqn.4.7, known as the Lorenz-Mie scattering theory [24]. In the limit of particles on the order of the size of the wavelength of light, the Rayleigh model is an easier, equivalent method derived from Mie theory.

To collect the light, there have been three main schemes used in literature; forward, side, and back-scattering microscopy schemes.

As shown in Fig. 7.7, the scattering distribution taken at different projections widely varies. In addition, the amount of light that gets collected in each of these length-scales also varies.

In the back-scattering scheme, the light collected is the sum of the incident light and the scattered light.

$$I(x,y) = |\vec{E}_r + \vec{E}_s|^2 \tag{4.14}$$

given that the incident wave is known as a polarized plane wave, this equation can be rewritten for the scattered light;

$$I = E_r E_r^* + E_s E_s^* + 2\mathrm{Re}[E_r E_s] \tag{4.15}$$

The incident field will be modulated by a scattering term, which is the Rayleigh scattering amplitude, S, as well as the reference term, R. Thus, the total intensity will be the sum of the scattering amplitude, reference amplitude, and the phase interference between the two fields;

$$I = |E_i|^2 \left[|R|^2 + |S|^2 + 2|R||S| \cos \Delta\varphi\right], \quad \Delta\varphi = \varphi_s \varphi_r \tag{4.16}$$

It should now be noted that the angle of image formation has not been specified. This is of importance as it allows to discuss the various methods of scattering microscopy without loss of generality between methods. Notably, the main difference is the angular content which is obtained from the various scattering models. In the case of spheres, this means using the various angular slices of the resultant scattered field.

4.5 Interferometric Light Scattering (iSCAT)

The key assumption in iSCAT is the relative magnitude of terms. The reference field is constant. The scattering term $R \approx k^4 \alpha^2 = \frac{V^2}{\lambda^4}$. Similarly, the inference term scales as the square root of this. Thus, the scattering term is dominated by the two fields, which we can ignore.

To accurately measure the positions, we can define the contrast metric, which relates the total field to that of a reference field;

$$\begin{aligned} C &= \frac{I - |E_r|^2}{|E_r|^2} \\ &= \frac{|S|^2 + 2|R||S| \cos \Delta\varphi}{|R|^2} \\ &\xrightarrow{|S| \ll |R|} \frac{2|S|}{|R|} \cos \Delta\varphi \end{aligned} \tag{4.17}$$

The phase of the interference consists of a sum of terms consisting of phases accumulated through the traversal of the scattered field. The total phase is dominated by the Gouy phase, which consists of phase accumulated when light is passed through a converging and diverging field, the de-focus phase, which can be parameterized to a relative phase shift, as well as the phase accumulated phase difference, which is typical of a Michaelson-interferomer type system.

4.6 Resolution limits of optical microscopy

The last thing to touch upon is understanding how the image that is being enlarged arrives to the camera, which consists of discussion of optical aberrations.

As mentioned in the formulation for the electric field scatter we have that any optical field consists of a set of plane waves that construct and deconstruct to form a field. That is, the electric field lies in the basis of a fourier series of terms of complex exponential functions. This fact allows one to describe the propagation of a wave in throughout z.

$$\vec{E}(x,y,z) = \iint_{\Omega} \vec{E}_0(k_x,k_y) e^{-i\left(xk_x+yk_y+\sqrt{k^2-k_x^2-k_y^2}\right)} dk_x dk_y, \quad \Omega = \{x \in \mathbb{R} : x^2 \leq k^2\} \tag{4.18}$$

We have that the energy of the wave is fixed by the trigonometric relation of the transverse wave vectors. Notably, this is fixed by the relation $|k| = \frac{2\pi}{\lambda}$.

The propagated wave passes through an objective which can be approximated by a lens. Using the ABCD matrix formulation, we can find that to have a clear image, the imaging condition must be satisfied, that is

$$\frac{1}{z_1} + \frac{1}{z_2} = \frac{1}{f} \tag{4.19}$$

where z_1, z_2 are the positions of the observer and the object, and f is the focal distance of the lens. Thus, the magnification of the lens is defined as $M = z_2/z_1$. However, to understand the aberrations it is necessary to have a field formulation of this problem. The simplest way to do this is described using linear optics theory.

Briefly, we define a pupil function $P(x,y)$ which is unitary if $r \leq R$, where R is the radius of lens. The point spread function of the object is then defined as a delta function through a two lens system. This is given by the fourier transform of the point source masked by the pupil function;

$$H(x,y) = \iint_{\Omega} P(x',y') e^{\frac{ik}{2}\left(\frac{1}{z_1}-\frac{1}{f_1}-\frac{1}{f_2}+\frac{1}{z_2}\right)\sqrt{x'^2+y'^2}} e^{-\frac{ik}{z_2}(xx'+yy')} dx'dy' \tag{4.20}$$

Expanding this, one can use a zero-order Hankel transform to integrate over the radius and find that the point-spread function, H, is defined in terms of the NA and the magnification;

$$H(r,z) = \int_0^1 e^{-\frac{ik}{2}\frac{\mathrm{NA}^2}{n}\rho^2 z} J_0\left(\frac{k\mathrm{NA}}{M}\rho r\right) \rho d\rho \tag{4.21}$$

where ρ is the rescaled radial vector $\rho = r'/R$, $\mathrm{NA} = nR/f_1$, and $M = nf_2/f_1$.

In perfect focus, we have that $z = 0$, for $z \neq 0$ we have a defocusing aberration which is defined as the secondary exponential. For a point source, which is diffraction limited, we find that the

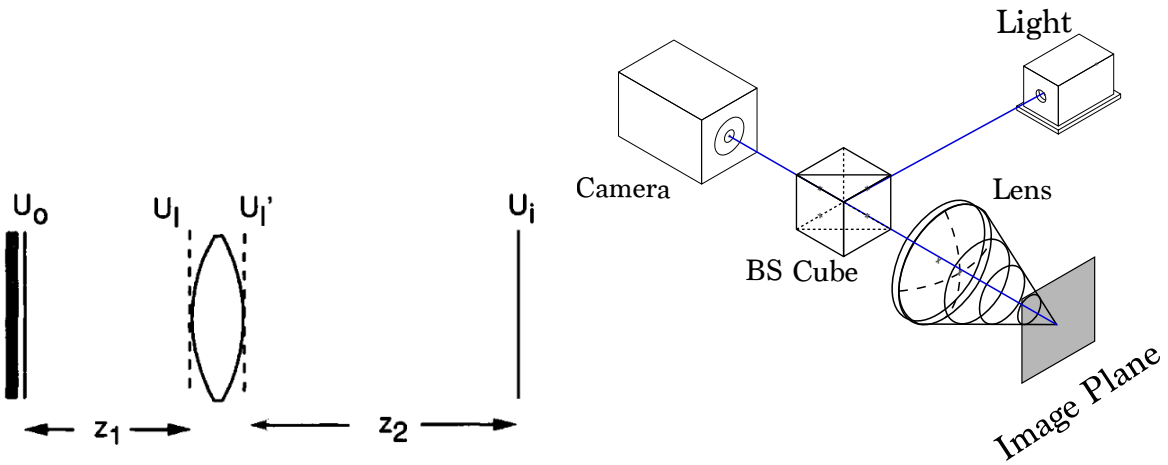


Figure 4.1: Simplest possible imaging setup. Image consists of a plane wave illumination, a lens, and a camera.

following is the limit of the optical microscope;

$$H(r,0) = |\frac{J_1(k\mathrm{NA}r)}{k\mathrm{NA}r}|^2 \tag{4.22}$$

We can solve that the minimum distance between the peak of the PSF and the minimum is solved by $k\mathrm{NA}r = x_0$ where x_0 is the first root of $J_1(x)$. This is classical known as the minimum transverse imaging criterion;

$$r_{\min} = 0.61\frac{\lambda}{\mathrm{NA}} \tag{4.23}$$

Similar results are found for the minimum axial imaging criterion by setting $r = 0$ and integrating.

$$z_{\min} = \frac{2\lambda n}{\mathrm{NA}^2} \tag{4.24}$$

At this point, we invite the fact that not all the light that scatters passes through the lens. Specifically, the range of k_x and k_y are fundamentally limited by the wavelength. The resolution limit is λ/NA. In the case of scattering microscopy, this is not a large problem as the scattering pattern is much larger, however, in the limit of two particle near each other, the spatial resolution limit must be enforced.

4.7 Sources of noise in imaging

In imaging, there is a number of sources of aberrations that one must account for. To discuss these sources of noise, a list of noises will be formed as one goes through the process of taking an image.

The schematic of an iSCAT microscope was already shown in Fig. 5.1, however, one can use a simplified model system without loss of generality. For a general imaging system, one only

requires a single lens for imaging. This simple setup is shown in Fig 4.1.

The lens will collects light from the object plane, U_o to the imaging plane, U_i, or the camera's plane. The propagation of that wave will collect aberration from the len's point spread function, as well as phase accumulation. The traversal over the point spread function is

$$U_i(x_i, y_i) = \iint_{\mathbb{R}^2} h(x_i, y_i, x_o, y_o) U_o(x_o, y_o) dx_o dy_o, \tag{4.25}$$

where $h(x_i, y_i, x_o, y_o)$ is known as the optical transfer function, or the response of a lens to a point-source, modelled as a delta-function. The len's magnification is incorporated into this using the following approximation;

$$h(x_i, y_i, x_o, y_o) \approx K\delta(x_i \pm Mx_o, y_i \pm My_o) \tag{4.26}$$

where K is a complex constant and M is a real-valued magnification. We can see that because of this, our image is spatially limited by this point-spread, which was earlier shown by the diffraction limit.

In addition to the spatial noise, there is a phase accumulation known as the Guoy phase. As the light, which can be assumed as consisting of an Eigen-expansion of plane waves, passes through a lens, these plane waves will converge to a gaussian beam with a beam-waist around the focal point. As this focusing occurs, there is a phase that is accumulated known as the Guoy phase. This phase accumulation is equal to

$$\psi(z) = \arctan\left(\frac{z}{z_R}\right), \quad z_r = \frac{\pi w_0^2 n}{\lambda}, \tag{4.27}$$

where w_0 is the waist of the final beam, n is the refractive index, and λ is the wavelength of light. In imaging, as long as the refractive index is constant, there is a total phase shift of π.

In the presence of a background beam, there will be a presence of a background. This background's source is the illumination beam itself, in the case of scattering microscopy. This background noise profile will take the shape of the illumination profile alongside the intensity of the power density.

The camera used will also contain electronic noise associated with it. For CMOS cameras, the noise present will consist of photon noise from the detectors and readout noise from the data-transfer. These noises can be modelled as Poisson noise from the photon detections and a gaussian for the readout noise.

In the case of using an imperfect lens, or a set of lenses as one has with an objective as one does in high magnification microscopy, there is a risk of spherical aberration. Spherical aberrations are caused by imperfections and misalignments in the lens itself. This cases the focal point of the lens to be slightly different depending on which area the light is focused by lens. For imaging

purposes, spherical aberrations cause various parts of the image to be out of focus, while others are in focus, creating non-uniform images.

Chapter 5

Interferometric Scattering Microscope Optical System

To observe single nanoparticle diffusion, a microscope system was designed. A scattering microscope is able to resolve nanoparticle three-dimensional position without a loss of signal over time, as presented in Chap. 4. The presented microscope optical system is able to track nanoparticles in three-dimensions by capturing a series of interference images with a CMOS camera.

5.1 Optical System Diagram

The iSCAT microscope system was designed based on prior work by the Kulkura lab REF. The iSCAT microscope is a wife-field optical system that illuminates the whole field of view with a coherent laser light. The iSCAT diagram was shown in Fig. 5.1.

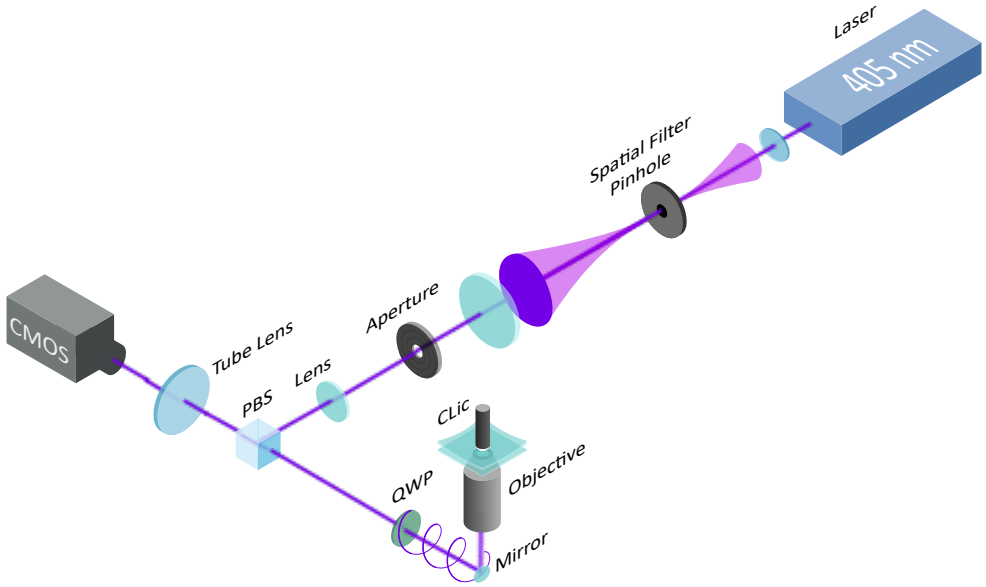


Figure 5.1: Isometric iSCAT Schematic.

The laser illumination is provided by a 405 nm laser with a beam-width of 1 mm. The laser beam travels through a $4f$-lens system, consisting of a spatial filter, to optimize beam shape, and a beam expander, to enlarge the beam diameter. The shaped beam is then passed through an aperture, which is used to adjust for various camera array sizes. The beam then is passed into a focusing lens, which directs it towards the objective. The beam first passes through a polarizing beam splitter, which partially reflects the linearly polarized light. The reflected light passes through a quarter-wave plate, which circularily polarizes te light. The circularily polarized light is then focused on the back plane of the objective. The objective focuses the light onto the sample plane, and the sample scatters the light back into the objective. The scattered light is then passed back through the quarter-wave plate, linearly polarizing it. The lgiht is then transmitted through the beam splitter and focused onto the CMOS camera.

The sample is contained between two glass cover-slips seperated by a plastic spacer. To control the gap between teh two coverslips, the convex-lens induced confinement (CLiC) system is used. The CLiC device is manually controlled with a control-box, which is not shown on the diagram. The control-box connects to a piezo controlled actuator and regulates the pushing force with fine precision.

5.2 Experimental Optical Table Overview

The laboratory implementation shown in Fig. 5.1, was assembled on an optical table. Fig. 5.2 and 5.3 show the lab setup.

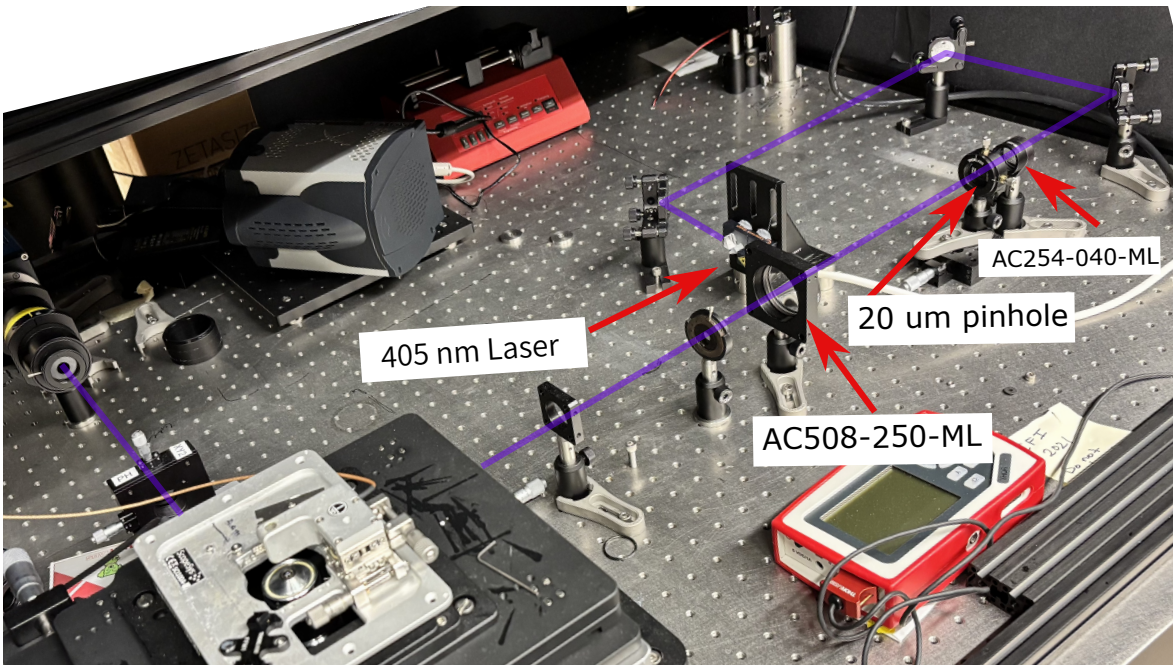


Figure 5.2: Clean table and retake the photos

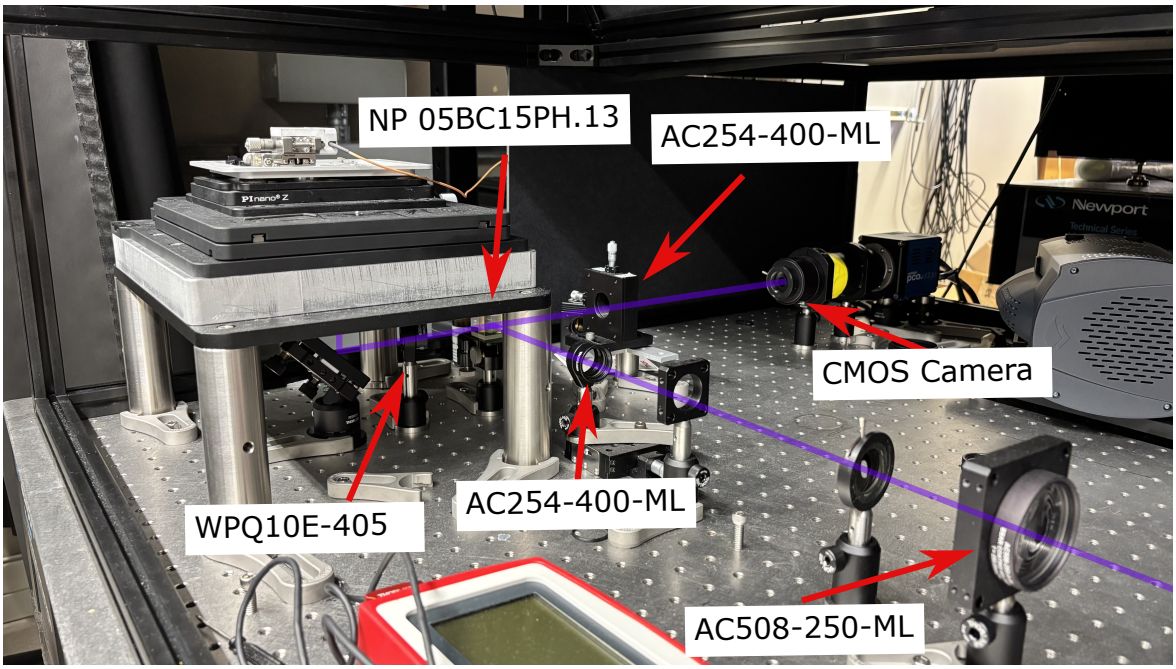


Figure 5.3: Clean table and retake the photos

The 405 nm laser, fabricated by Coherent, was controlled by a Thorlabs laser driver, which varies the output power. The laser driver was set to 850 mW which outputted a laser light intensity of

$\sim 220\,\mathrm{mW}$.The Thorlabs power meter was able to measure the output light intensity with high accuracy. The laser driver is shown in Fig. 5.4.

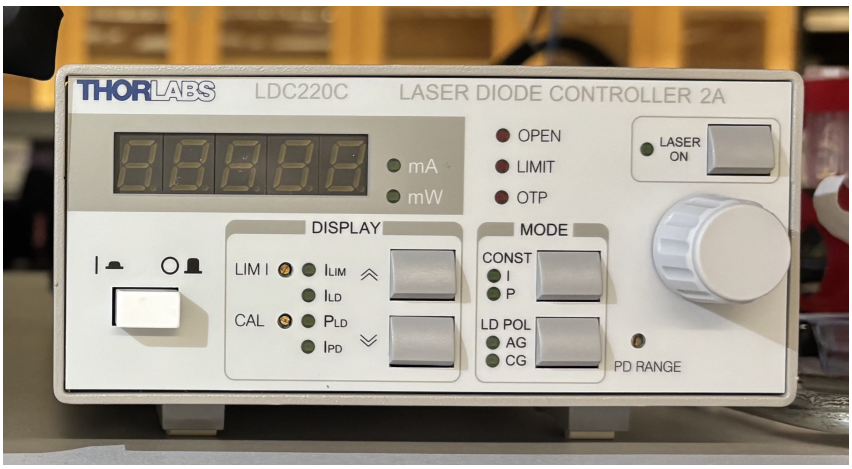


Figure 5.4: Laser Driver

Imaging was performced with an oil-immersion objective with magnification factor $60\times$ and numerical aperture NA = 1.4. The immersion oil was used required to match the refractive-index of the coverglass to the objective. The tube lens, located in front of the CMOS camera, was selected with a magnification factor of $2\times$. The total optical magnification factor was $120\times$. The final image was digitally zoomed such that each pixel represents $55 \times 55\,\mathrm{nm}$. Since the camera pixel array is $6.5\,\mu\mathrm{m}$ in size, it produces a total magnification of $118\times$.

The table in Tab. **??** shows the iSCAT microscope components list. The components list are divided into components for the laser head, spatial filter, objective focusing, and camera equipment. All mirrors have the same part number and manufactured by Thorlabs.

The camera used was the PCO.EDGE 4.2. This camera has a pixel array of 2048×2048 with pixel size $6.5 \times 6.5\,\mu\mathrm{m}$. The camera can produce images of the full field of view at a maximum rate of 40 fps. When a smaller field of view is used, the camera can reach up to 1000 fps. The camera saves the images with the timestamps to provide precise framerates.

5.3

Component	**Product ID**	**Manufacturer**
Laser head	—	—
Mirror	BB1-E02P	Thorlabs
SF Lens 1 ($f = 40\,\mathrm{mm}$)	AC254-040-A-ML	Thorlabs
SF Pinhole ($20\,\mu\mathrm{m}$)	P20K	Thorlabs
SF Lens 2 ($f = 250\,\mathrm{mm}$)	AC508-250-A-ML	Thorlabs
Objective focusing lens	BB1-E02P	Thorlabs
Polarising beam splitter	05BC15PH.13	Newport
Beam splitter mount	9411	New Focus
Quarter-wave plate	WPQ10E-405	Thorlabs
60× oil-immersion objective	—	Nikon
Tube lens ($f = 40\,\mathrm{mm}$)	AC254-040-A-ML	Thorlabs
sCMOS camera	PCO.EDGE 4.2	Excelitas

Table 5.1: Optical components of the iSCAT microscope.

Chapter 6

Simulations of Diffusion of Nanoparticles in Double-Layer Walls

Langevin dynamics are a framework for computational simualations that model the model of particles by accounting for both random thermal fluctuations arising from surrounding environment and deterministic forces such as external fields or interactions with other particles.

This chapter descirbes a simulation model of nanoparticle brownian motion within a confined space subject to electrostatic field generated by two parallel charged surfaces.

Python software was written based on prior Langevin python implementation published on Github [1]. Langevin algorithms were implemented and parameterized to account for physical electrostatic forces and charge quantities.

Using the software, statistical averages of the particle motion was calculating for a range of electrostatic screenings length presented in the sections below.

6.1 Simulation Equations

The general Langevin equation is generally written as

$$m\ddot{\vec{x}}(t) = -\lambda\dot{\vec{x}}(t) + \vec{F}(t) + \Gamma\vec{W}(t), \tag{6.1}$$

where λ is a velocity dependent friction coefficient, Γ is the magnitude of fluctuation, $\vec{F}(t)$ is a deterministic force, and $\vec{W}(t)$ is a vector random variable.

The Langevin equation programmed into the software is described as

$$m\ddot{\vec{x}}(t) = -\gamma(z)\dot{\vec{x}}(t) + \vec{F}(\vec{x}(t)) + \sqrt{2k_bT\gamma(z)}\vec{W}(t), \tag{6.2}$$

where physical magnitudes are implemented; $\gamma(z)$ is an axially dependent hydrodynamic friction depending on how close the particle is to a surface. The diffusion coefficient in an axially dependent friction is writte using the Stokes-Einstein relation,

$$D(z) = \frac{k_b T}{\gamma(z)}. \tag{6.3}$$

In over-damped case, where $m\ddot{\vec{x}} \ll 1$, the the equations of motion can be written as

$$\begin{aligned} \frac{dx}{dt} &= \sqrt{2D_x(z)}W_x(t) \\ \frac{dz}{dt} &= \sqrt{2D_z(z)}W_z(t) + \frac{F_z(z)}{\gamma(z)}, \end{aligned} \tag{6.4}$$

as described in Eqn. 7 of Ref. [1]. This model is especially valid for particles of diameter between $10\,\text{nm} - 100\,\mu\text{m}$. This model is valid for particle such as PMMA, carboxylate-modified polystyrene, and similar materials.

Since $\vec{W}(t)$ is a random, non-analytical function, classical ODE methods of time evolution do not apply. To simulate the stochastic portion, the Euler-Maruyama method should be used.

Eqn. 6.4 can be written into the Itô form,

$$\begin{aligned} dX_t &= \sqrt{2D_x(z)}dW_x \\ dZ_t &= \sqrt{2D_z(z)}dW_z + \frac{F_z(z)}{\gamma(z)}dt, \end{aligned} \tag{6.5}$$

where stochastic quantities are indicated in capital letters.

In the Itô form, the Euler-Maruyama equations are written as a pair of discrete time-step evolutions. Each equation described a random realization of the position, $\vec{x}(t)$, that depends on the random function, $\vec{W}(t)$. The equations are programmed into the software, referenced to as MJ_LANGEVINSIMULATIONS,

$$\begin{aligned} x_{n+1} &= x_n + \sqrt{2D_x(z_n)}W_{x,n}\Delta t \\ z_{n+1} &= z_n + \left[D_z'(z_n) - \beta D_z(z_n)V'(z_n) + \sqrt{2D_z(z_n)}W_{x,n}\right]\Delta t, \end{aligned} \tag{6.6}$$

where $V(z)$ is the electrostatic potential, $\beta = 1/k_b T$, and Δt is the discretized time.

The software accepts parameters inputted from a generic config file and save results as a HDF5 file using the Python library H5PY.

6.2 Python Simulation Implementation

Opis software

6.3 Electrostatic Input Parameters

The deterministic portion of the Langevin equation in the simulations are determined by the electrostatic and grativational forces acting on the particle. In the model of nanoparticles, the grativational forces are negligible, leaving the electrostatic forces as the driving non-spurious drift. A diagram of the simulation geometry is shown in Fig. 6.1.

$$a = 50\,\text{nm},\ L = 300\,\text{nm},\ \lambda_D = 40\,\text{nm},\ B = 25\,\text{mV}$$

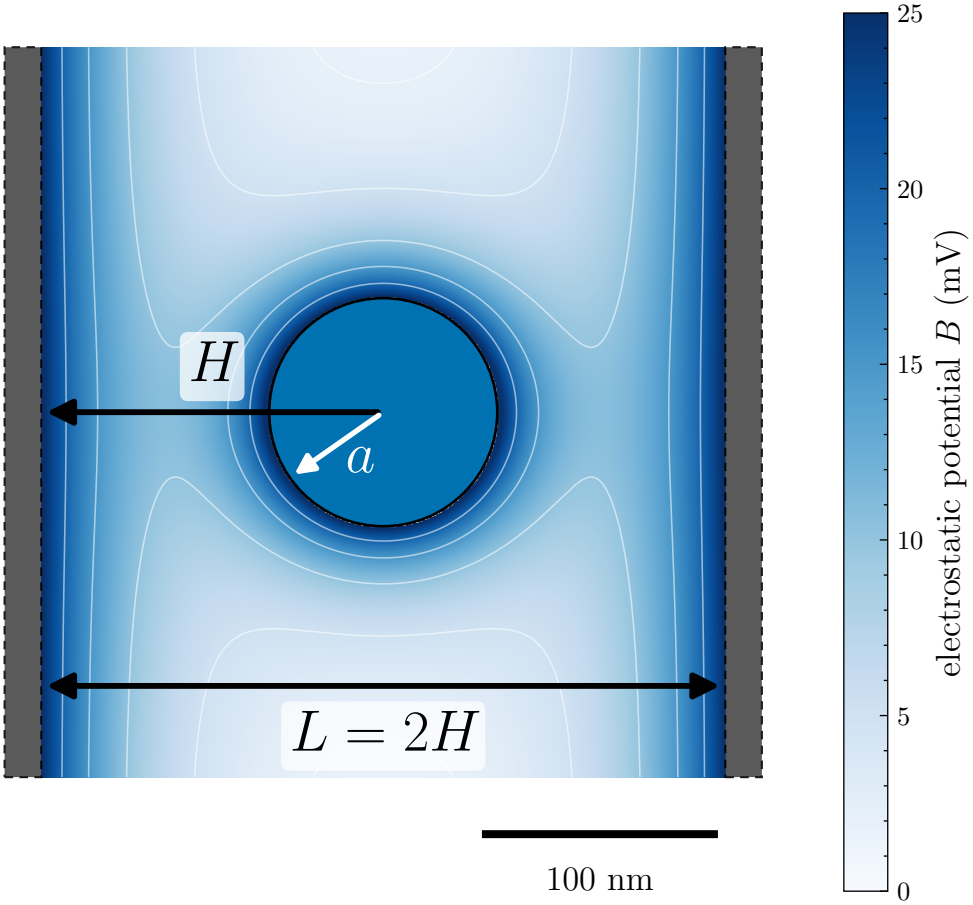


Figure 6.1: Model diagram of simulation geometry. A spherical particle of radius a is modelled between two charged parallel plates.

The electrostatic forces that drive the dynamic motion are described by the following equations,

$$U(z) = B\left[e^{-(H-z)/\lambda_D} - e^{-(H+z)/\lambda_D}\right], \tag{6.7}$$

where the pre-factor, B, is described by the function $f : (\zeta_p, \sigma_w, a, \lambda_D, T)$. To ensure that the equations represents the sample of interest, the electrostatic amplitude is calculated from

laboratory measureable quantities. The function is described the by the Grahame model,

$$B = C \cdot \tanh\left(\frac{e_0\zeta_p}{4k_bT}\right) \cdot \tanh\left(\frac{e_0\psi_w}{4k_bT}\right) \tag{6.8}$$

$$C = \frac{64\pi\varepsilon_0\varepsilon_r a k_b T}{e_0^2} \tag{6.9}$$

$$\psi_w = \frac{2k_bT}{e_0}\text{asinh}\left[\frac{\sigma_w e_0 \lambda_D}{2\varepsilon_0\varepsilon_r k_b T}\right], \tag{6.10}$$

where ζ_p is the zeta-potential of the particle, ψ_w is the stern-layer potential for the surface walls, a is the particle radius, e_0 is the elementary charge value, and λ_D is the Debye-length of the solution. These equations are physically constrain the electrostatic field amplitude to ensure physical motion during the simulation process.

6.4 Mean-Squared Displacement in Confinement

The simulation results provide three-dimensional trajectories of particles. From the trajectories, the mean-squared displacement can be calculated. The typical output of a simulation is shown in Fig. 6.2. The plot on the left depicts the transverse motion, $X(t)$. The plot on the right depicts the axial motion with respect to the confined walls. The plot in Fig. 6.2.

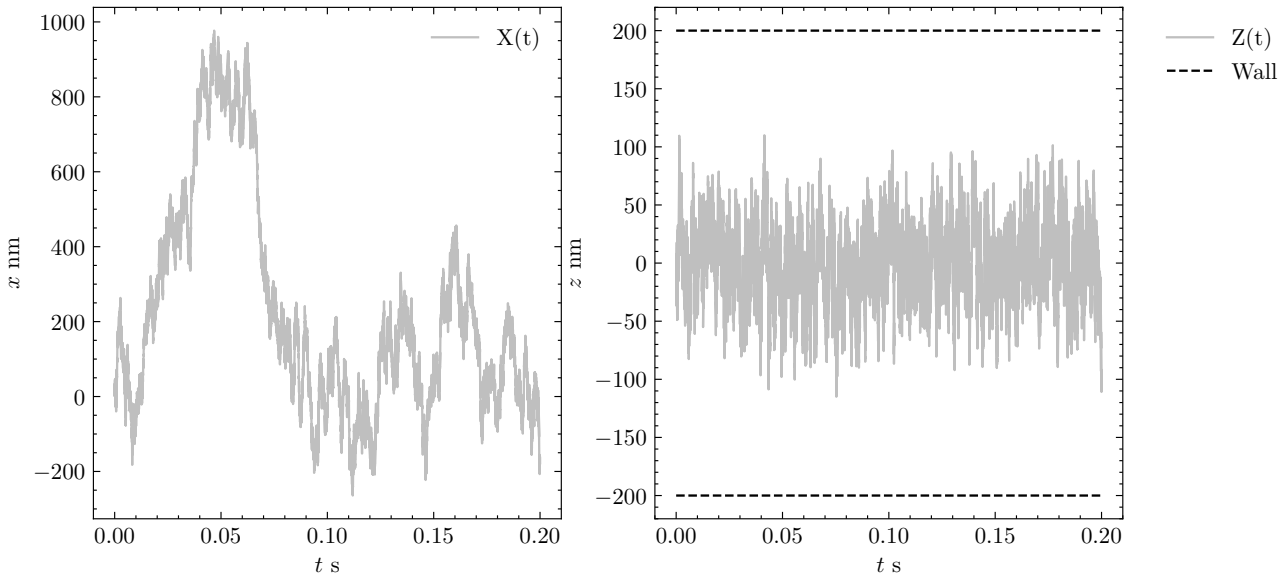
For each simulated particle trajectory, the MSD can be found. The results are statistically inferred parameters that can be compared to the baseline inputs.

The size of a particle can be measured from the mean-squared displacement. In confinement, an addition correction must be applied due to the axial friction. In the case an electrostatic field, this correction is not applicable due to the fact that the not known.

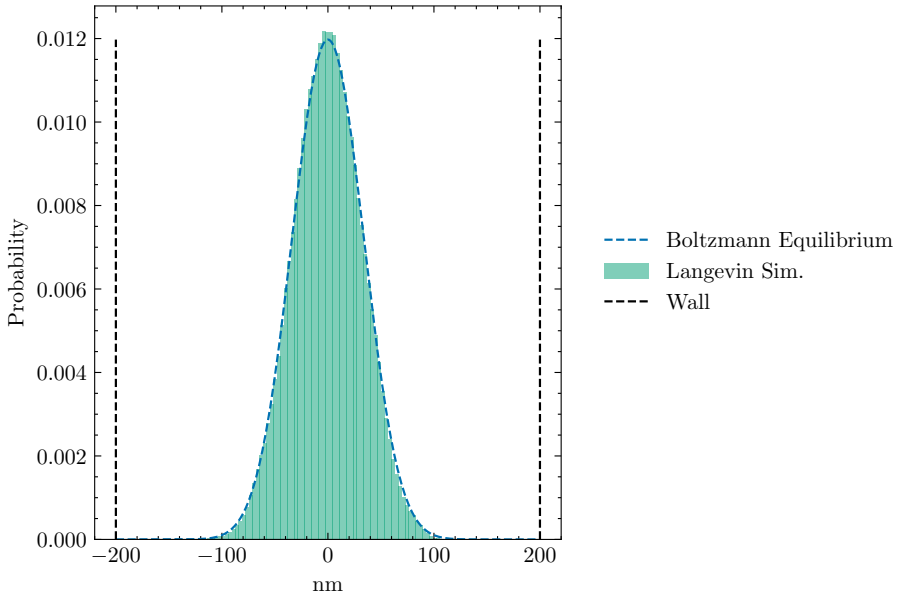
To quantify the confinement length changes as a function of the electrostatic field, the MSD of the axial and transverse trajectories are compared in two cases. One case is a highly negatively charged paricle, and the second for a neutrally charged particle. This is shown in Fig. 6.3.

The simulations demonstrated that as the particle charge becomes greater in magnitude, there is a large apparent change in the MSD plateau. This phenomena is expected, as the electrostatic field focuses the particle into the center of the two plates. This relationship is a function of the zeta-potential of the particle and can be explained using Coulomb's law as explained in Chap. 2.

The effects of the electrostatic field interaction on the confinement length can be investigated by solving the mathematical relations analytically, or explored using simulations developed for this purpose. The theoretical MSD function can be mathematically determined if one solves the Fokker-Planck equation in the presence of a Debye-Hückel potential. This combination of equations in confinement does not have an analytical solution. Thus, simulation approaches must



(a) Trajectories of transverse and axial motion during single realization of Langevin simulations.



(b) Histogram of position over time.

Figure 6.2: Confined Brownian Particle realization of z diffusion in electrostatic potential. The step-histogram is plotted in green. Simulation input parameters are $a = 50\,\text{nm}, \zeta_p = -3.5\,\text{mV}, \sigma_w = -2\,\text{mC/m}^2, L = 225\,\text{nm}, \lambda_D = 100\,\text{nm}, T = 300\,\text{K}, \eta = 1\,\text{mPa s}$.

be taken.

6.5 Simulation Results

Using the developed software, the relationship between the effective confinement length can be numerically determined as the software numerically approximates the system. In order to find correlation to laboratory experiment, the electrostatic field must be varied as the zeta-potential of the particle is constant. In the simulation, the electrostatic field can be easily manipulated

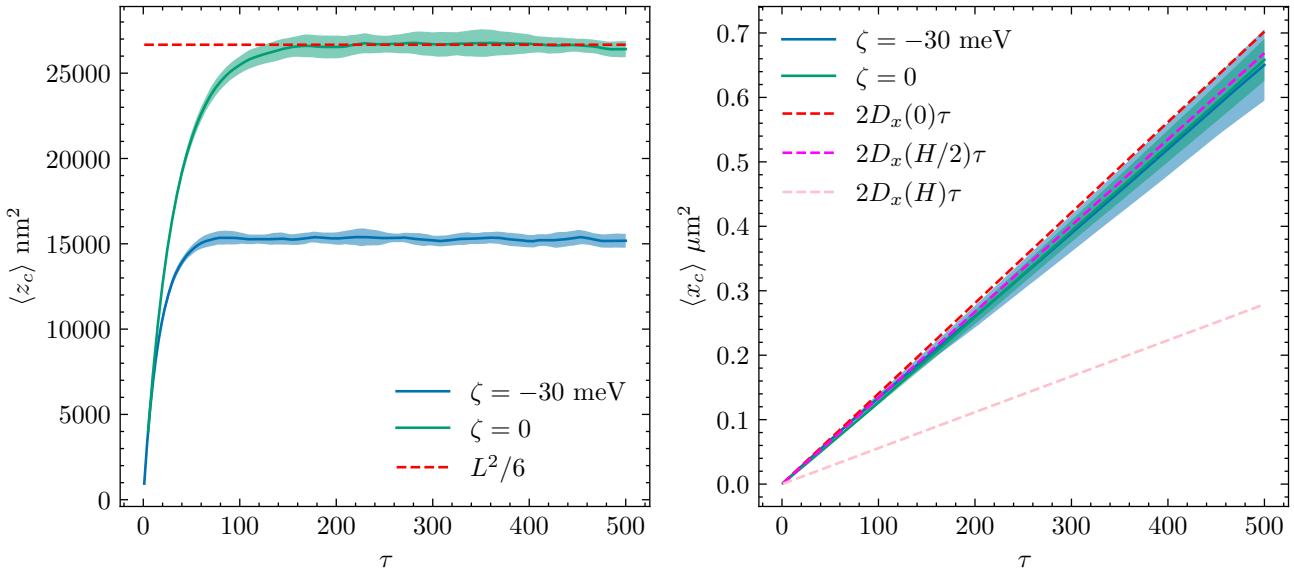


Figure 6.3: Comparison between MSD curves of two different charged particles. The green trace represents a neutrally charged particle with a maximal MSD of $\sim 26\,000\,\mathrm{nm}^2$. The blue trace represents a negatively charged particle with a maximal MSD of $\sim 15\,000\,\mathrm{nm}^2$. The chart on the right shows the transverse MSD of the two particles. Exposure time $\tau = 200\,\mu\mathrm{s}$, $\lambda_D = 10\,\mathrm{nm}$.

and in the laboratory experiment, same can be achieved by varying the ionic concentration of the solution. Changing the electrostatic amplitude in laboratory experiments is possible as changing the Debye-length is inversely proportional to the changing, ionic concentration;

$$\lambda_D \propto \frac{1}{\sqrt{C}}, \tag{6.11}$$

where C is the ionic concentration of a dilute electrolyte compound REF.

The simulation for the geometry described above was conducted for various Debye-lengths. The conditions of the simulation are such that the zeta potential of the particle and glass are chosen to resemble glass and a polystyrene bead. The particle parameters were chosen to be $a = 50\,\mathrm{nm}, \zeta = -30\,\mathrm{mV}$. The geometric conditions are: $\sigma_w = -2\,\mathrm{mC/m}^2, L = 500\,\mathrm{nm}, T = 300\,\mathrm{K}, \Delta t_i = 0.2\,\mu\mathrm{s}, \tau = 200\,\mu\mathrm{s}, \eta = 1\,\mathrm{mPa\,s}$.

The simulation results are shown in Fig. 6.4

The figure obtained from the simulation shows that as the Debye-length increases, the confinement length decreases. When the ion concentration increases, the particle is less screened which allows it to move closer to the walls. With increased concentration of the ions, the particles can more easily explore the volume. Once the concentration reaches a sufficient saturation, the particle move from wall to wall, or a distance of L. As the ionic concentration approaches 0, the confinement length approaches a minimum of $\sim 50\,\mathrm{nm}$. Theoretically, this should reach 0 nm, however, due to simulation artifacts it doesn't. The extra motion is attributed to having a finite time-step during the evolution of the Langevin equation, adding extra force. In the limit of an

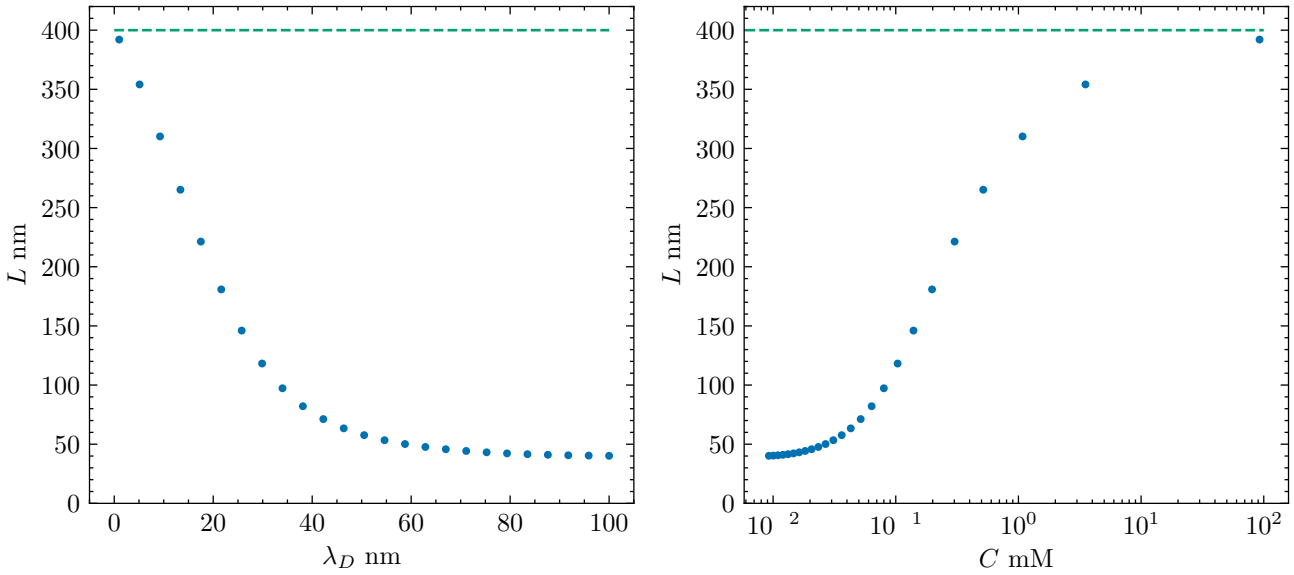


Figure 6.4: Effective Confinement length plotted against the solution's Debye-length.

infinite time step, the minimum confinement should approach 0 nm.

As a result from the changing effective confinement length, the diffusion coefficient changes as a function of ionic concentration. Since diffusion is dependent on the axial position of the particle with respect to the wall, there is a relative diffusion decrease with addition of ions. The exact relationship can be investigated using the developed simulations.

Using the simulated trajectories in Fig. 6.4, the diffusion coefficient of the particle is be measured from the first 15 points of the MSD slope by fitting to a line as performed in optical experiments. The resulting diffusion coefficient is then compared to the baseline input parameter. This is plotted in Fig. 6.5.

The above results that the presented simulations can be used to distinguish between classes of particles. This is because the relationship between the effective confinement length and the relationship between diffusion and ion concentration is established. There is a notable 10% drop in the diffusion coefficient at high concentrations of ions. This is attributed to the particle being closer to the walls for a longer peroid of time.

By doing more simulation cases, this relationship can be further investigated.

The effective confinement length compared to the zeta-potential of a particle at various Debye-lengths is simulated in the same geometric conditions as Fig. 6.4. This is plotted in Fig. 6.6.

Fig. 6.6 presents the relationship between L, ζ, and the Debye-length, λ_D. Each curve shows the relationship between the values for a different ionic concentration.

At high concentration, there is little to no effect on the confinement length, while at low

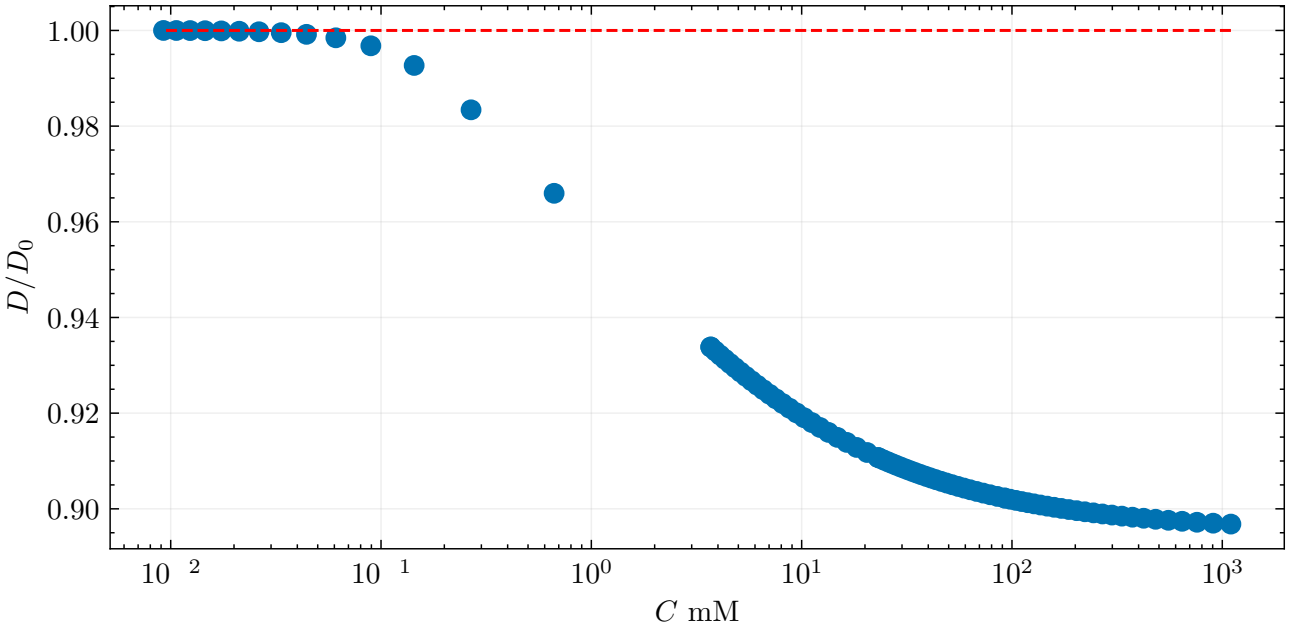


Figure 6.5: Effective diffusion plotted against the solution's Debye length.

concentration, there is a steep dependency near $\zeta \to 0$. As the particle becomes more negatively charged, the confinement length decreases towards the confinement length minima. When the particle becomes more neutral, the confinement length approaches the gap length between the two walls.

6.6 Surface Charge estimation

The surface charge of the confinement walls can be estimated based on the electrostatic potential. Since the electrostatic force determines the drift coefficient, the drift amplitude can be statistically recovered using a maximum likelihood estimator function.

Although the SDE MLE equation is already known, the solution was specifically applied for the purpose of this thesis in the form presented in Eqn. 6.12. A detailed derivation is presented in Appendix. A;

$$\hat{B} = \frac{\sum_n \frac{\phi(z)}{2D_z}\left[\Delta z - D'(z)\Delta t\right]}{\sum_n \frac{\phi^2(z)\Delta t}{2D_z}}, \quad \mathbb{V}[\hat{B}] = \sum_{n=0}^{\infty} \frac{\phi^2(z)\Delta t}{2D(z)}, \tag{6.12}$$

where Δz is the random variable assigned to step displacement, $D(z)$ is the diffusion coefficient, and $\phi(z)$ is the gradient of the potential,

$$\phi(z) = e^{-\frac{H}{\lambda_D}} \sinh\left(\frac{z}{\lambda_D}\right). \tag{6.13}$$

This estimator assumes that gravitational contributions are negligible.

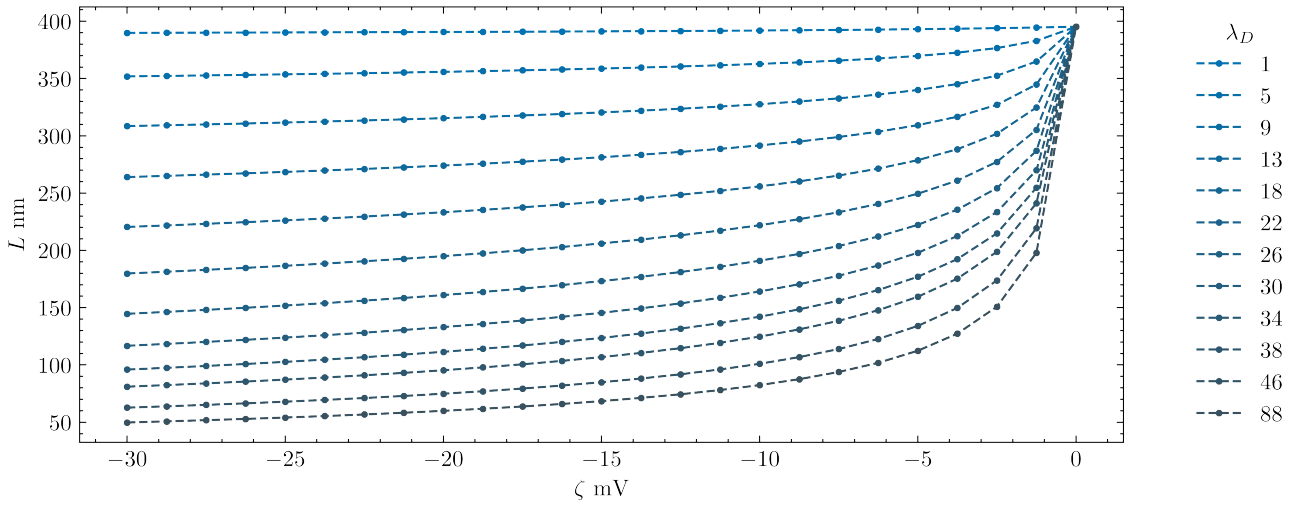


Figure 6.6: Effective confinement plotted against the particle's zeta-potential.

If the amplitude of the electrostatic interactions is known, one can invert the amplitude using relations in Eqn. 6.8. This inversion requires at least the zeta-potential, ζ, of the particle, the radius, a, and Debye-length, λ_D.

The particle's exploration, demonstrated in Fig. 6.2, showed a clear dependence on a drift due to the electrostatics. This drift is a useful estimator of the particle-wall charge relationship. Using the aforementioned amplitude maximum-likelihood estimator function, the drift estimator can be extracted into meaningful, physical information.

To determine the effective accuracy of the MLE function, simulations are performed using the developed software.

The Langevin simulation was ran for 20 ms with time-step of 0.2 µs. In this histogram there are 50 simulations with seeds set to $S_{\text{eed}} = i * 10 + 1$ for $i \in \{0, 49\}$ using the *numpy* seed parameter. $a = 50\,\text{nm}, L = 500\,\text{nm}, T = 300\,\text{K}, \Delta t_i = 0.2\,\mu\text{s}, \tau = 200\,\mu\text{s}, \eta = 1\,\text{mPa}\,\text{s}, \lambda_D = 10\,\text{nm}$ The surface charge is fixed at $\sigma_w = -2\,\text{mC/m}^2$. The distribution of the MLE is shown in Fig 6.7.

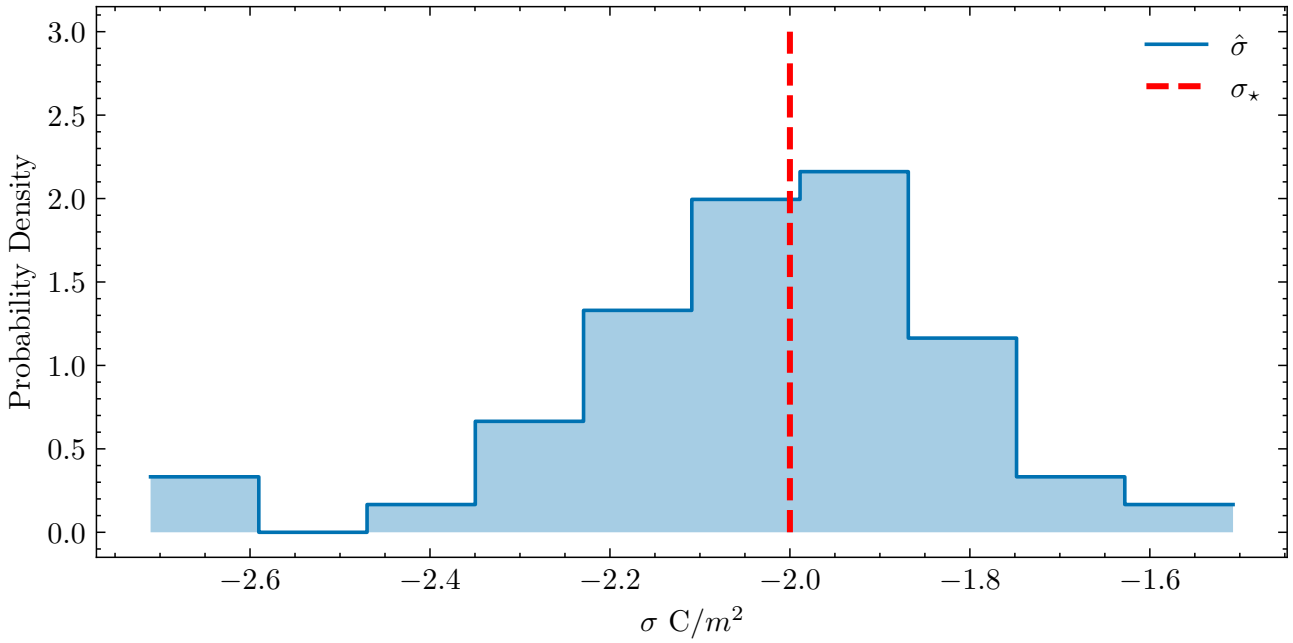


Figure 6.7: Distribution of $\hat{B}$ Re-parameterized for the surface charge. $\hat{\sigma} = 1/(A \cdot C)^2 \hat{B}$ where A, C are properly defined in 6.8.

Chapter 7

iSCAT z-estimation Simulations

This section presents a computational method for determining the axial (z-axis) position of a nanoparticle from iSCAT imaging data. The proposed approach is based on principles of light-scattering in the Rayleigh approximation, which govern the inference pattern as the nanoparticle moves along the optical axis.

In order to evaluate the accuracy and performance of the proposed method, Mie-theory simulations are used for for determining the an ideal nanoparticle's z-axis position. The precision and limitations are discussed in the context of recovering three-dimensional particle trajectories.

The raw images are acquired from a CMOS camera according the optical schematic illustrated in Figures 5.1 and 8.1. Each image consists of a two-dimensional interference recorded at a specific focal position z_{focus}, defined as the distance between the objective's focal distance and the objects in the field of view.

7.1 Method Overview

The method described in this section is based on computational image processing on images acquired using a CMOS camera. The camera has an array of 2048×2048 pixels. To acquire images at higher frame rates, only a portion of the CMOS array is used. This is set during the experimental details. Typical sizes used are 600×450 at 400 fps however, other fields of view can be used for specific experimental purposes. The highest frame-rate possible with this camera is 1000 fps at a FOV of 200×250.

The object sample is imaged through an oil-immersion objective lens. The optical system provides a magnification of $60\times$, with a tube lens of magnification $2\times$. The remaining is an digital zoom with a final resolution of 55 nm per pixel. The final magnification is then the ratio of the CMOS array length divided by the pixel size.

The image-processing workflow used to determine three-dimensional trajectory is shown in 7.1.

The diagram summarized the sequence of image processing steps to extract each coordinate position.

1. **Acquisition** — raw image captured on camera.
2. **Background subtraction** — conversion of raw image to an iSCAT contrast image.
3. **Lateral tracking** — transverse particle position, (x, y), located in each frame.
4. **Angular integration** — extraction of the scattering data per frame.
5. **Nuisance-parameter estimation** — Fix iSCAT Amplitude, microscope decay parameter.
6. **Linking** — axial candidates, $z_{\min}$, joined by the 2D Viterbi algorithm.
7. **Output** — the three-dimensional trajectory $\big[x(t), y(t), z(t)\big]$.

Figure 7.1: Outline of the 3D particle-tracking pipeline.

7.2 Raw image background subtraction

To obtain high-contrast interference signal from raw camera data, a post-processing background subtraction routine is introduced. This background subtraction measured a temporally averaged background using a window-filter and Fourier filtering. An example of this process is shown in Fig.7.2.

The full algorithm is presented in Alg. 1. The overview of the algorithm is to subtract the background noise in order to minimize Poisson, and electronic noise.

To reduce the contribution of the static background, the mean background is subtracted from each frame. Since the background illumination originates from the gaussian illumination beam, which has a constant mean intensity over the acquisition period, this stationary background removes the background offset while preserving dynamic iSCAT signal. This just requires a good estimation of the background in order not to remove any iSCAT signal. The remaining shot noise, which follows Poisson statistics is reduced by temporal averaging during background estimation.

In order to suppress electronic noise from data-transfer, a row-wise averaging scheme is applied. Row-wise means are taken and normalized with respect to their row as the data-transfer from CMOS to computer is by horizontal lines. Thus, to ensure all lines are equally weighted, they are normalized with respect to their entire row. This is implemented as per prior iSCAT normalization methods by the Sandoghar laboratory Vahid's github should be here.

The set of images, which consist of matrices of pixel-values of shape $F \times H \times W$, or F number of frames, W pixels wide, H pixels tall.

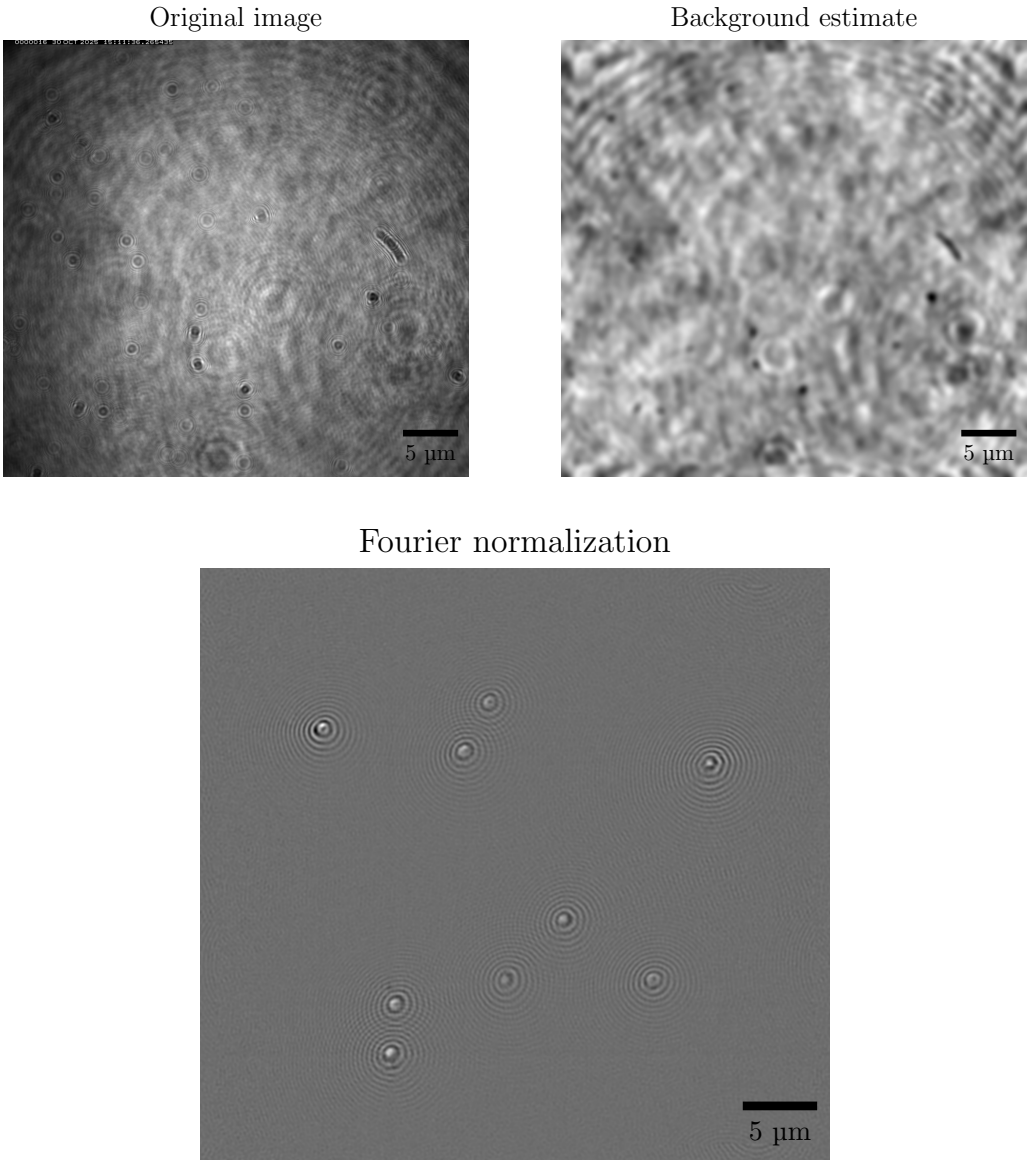


Figure 7.2: Raw image captured on camera to background subtracted iSCAT image. Single particles are tracked and extracted for further analysis.

Before processing begins, the parameters of the Fourier-band-pass filter are determined from the image dimensions. The filter consists of an annular, donut mask of inner radius equal to 1% if the larger image dimension and outer radius equal to 25% of the larger image dimension, corresponding to $r_i = \max(H, W)/100, r_o = \max(H, W)/4$.

To minimize edge artifacts, a binary mask is applied to each frame. Pixels within 20 pixels of the boundary are assigned a value of 0 using a mask filter. On the interior pixels, the mask is set to 1, while on the pixels less than 20 pixels away from the edge are set to 0. This mask is used during normalization to prevent edge effects from biasing the inner statistics.

For each frame in the stack, the image is first normalized using a row-wise normalization function. The function, titled AmpNorm in Alg. 1, first obtains the masked-image. The global normalized

mean is stored. For each row, the mean is taken and the row of the global-mean is adjusted by the row-wise mean. The final image is the global mean divided by the row-mean for each row.

Once normalized, the background of the image is calculated by taking the forward and backward ΔL frames spaced by a spacer of Δf frames and taking the AmpNorm of these frames.

The iSCAT image then consists of a background subtraction between the normalized image and the background. Finally, once background subtracted, the image is Fourier-filtered using the pre-allocated band-pass filter. This process is demonstrated in Fig. 7.2

7.3 iSCAT (x, y) Tracking

To determine the transverse positions of any particles detected in the iSCAT method, the radial variance transform algorithm is used [26]. The original algorithm is briefly outlined below for completeness. The full code is written by the Sandoghdar group and can be found in Ref [26]. A demonstration of this method is presented in the Fig. 7.3.

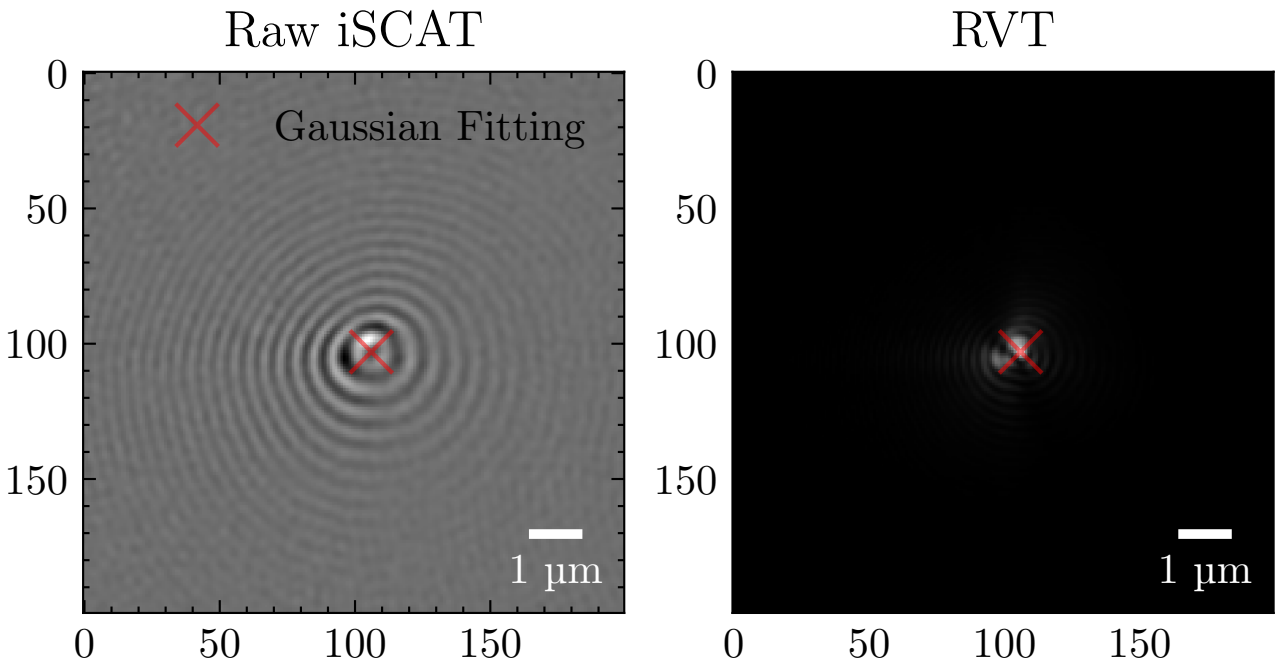


Figure 7.3: Raw image to RVT. Gaussian fitting of the RVT is shown on both plots.

The transverse localization algorithm, referenced as TransverseLocalization in 2 consists of transformations onto the iSCAT image from the previous section.

The input image stack is first loaded and assumed to consist of radially symmetric interference fringes. The image is first transformed using the Radial Variance Transform [26]. This function converts a radial-fringe pattern into a point-like delta function when a radial symmetry is acted upon it.

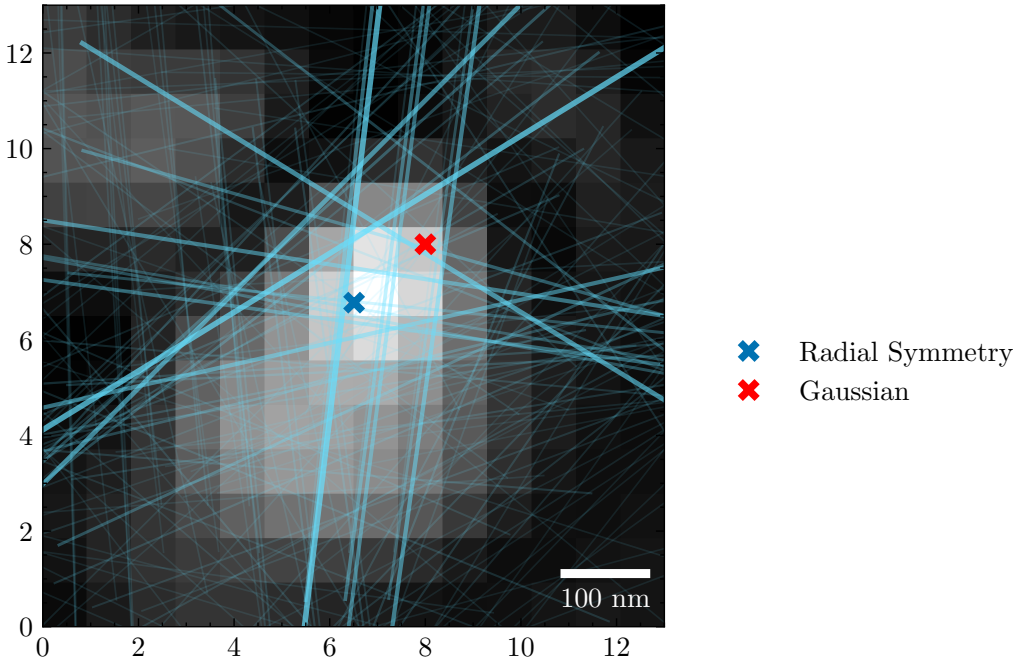


Figure 7.4: Demonstration of the Radial Symmetry method applied onto RVT of iSCAT data. Implemented from [27]. Figure adapted from [27].

Before proceeding, a frame-independent parameter for threshold SNR amplitude, number of pixels, and ROI cropping parameters are inputted. for a particle to be detected, the SNR of the RVT should be roughly 90% of the max iSCAT amplitude. An estimate of particle number is assumed, which acts as a prior for top n detections. Lastly, ROI for sub-pixel estimation is chosen.

For each frame, the iSCAT image is dilated. Particle positions that are considered maxima if the dilated image is equal to the RVT and the RVT is above the threshold for detection.

Once candidates are chosen, a NL-LS gaussian fitting to the RVT is performed to acquire higher-accuracy localization. To improve this to sub-pixel, a crop is done around each RVT gaussian fitted point and a radial-symmetry algorithm is performed as per [27].

To ensure accurate linking, an implementation of the Munkres algorithm is used REF. Briefly, a cost-matrix of distances is constructed and minimized in time to ensure that optimal displacements are chosen. This algorithm has limitations in cases where particles overlap. Ideally, experimental measure are taken to minimize this occurring. One such measure is to lower the concentration of particle in the recorded images.

7.4 Angular Integral

Image data will have background misalignment due to iSCAT hardware setup. To optimize for fitting speed and misalignment property, radial scattering profile are assumed as in the Rayleigh

regime model is symmetric. The radial integration method is described in Alg 3.

The discrete angular integral consists of summing angular terms on a radial grid. Mathematically, the output array obtained is described by the relation

$$g_t(\rho) \tilde{-} \frac{1}{\Delta\theta} \int_{\theta_0}^{\theta_0+\Delta\theta} I_t(\rho,\theta) d\theta, \tag{7.1}$$

where θ_0 is a constant, $\Delta\theta$ is the width of integration, $I_t(\rho,\theta)$ is the iSCAT image.

First, the cartesian image is Re-parameterized into polar coordinates using a change of basis transformation on the index grid of the image. Then, for each ring of indices, the image coordinates are summed up on this ring. The final array is an angular average that is indexed at radius r for some upper radial bound. This process is illustrated in Fig. 7.5.

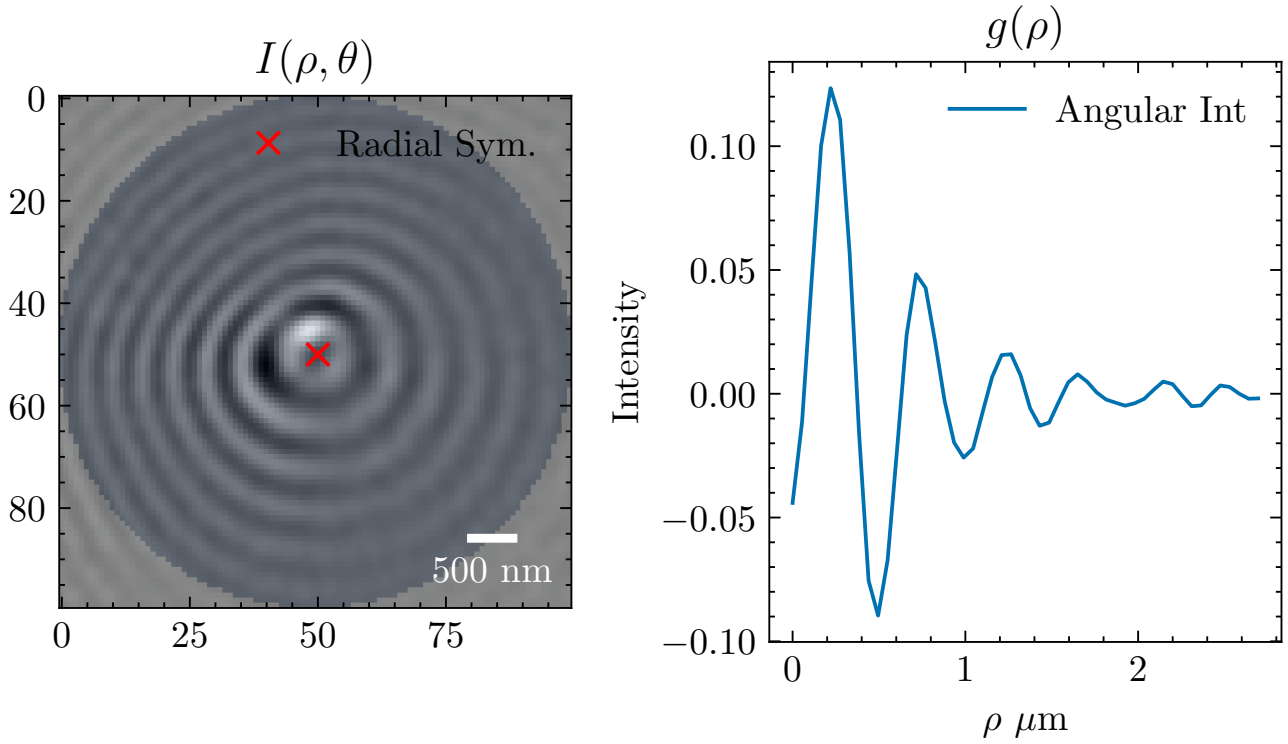


Figure 7.5: Angular Integral demonstration.

7.5 Nuisance parameter Estimation

In order to speed up computation and produce a more statistically relevant result, the method aims to fix the model parameters which should theoretically stay constant. These, for example, are the particle mass signal and optical aberration coefficients. These parameters, known as nuisance parameters, should be fixed before fitting to the more sensitive parameter of interest, the relative z position.

To obtain accurate estimation of the z position relative to nearby frames, nuisance estimation is done to focus on z fitting. To do this, L-BFGS-B Fitting is performed. alongside amplitude fitting. The theory of nuisance parameter estimation is described in [28].

The nuisance parameter method works on the premise of a model function. The iPSF model described in 4 is used as a weight function to compare to the data. First, bounds are placed for physical values of each parameter.

Here, it is important to note that the amplitude of the signal is proportional to the mass. The spatial decay of the aberration is modeled as a decaying exponential.

For each candidate decay parameter, the weighted correlation function is calculated. Let g be the radial profile, and G the theoretical iPSF model. The inner product of these two arrays, weighted in some integral of radiuses, is defined by $\langle G, g\rangle_w$. The correlation function is then written as

$$\rho(z,\gamma) = \frac{\langle G, g\rangle_w}{\sqrt{\langle G, G\rangle_w + \langle g, g\rangle_w + \varepsilon}}, \tag{7.2}$$

where ε is a small number to prevent division by 0.

The best-fit amplitude function here is analytical, given by the expression

$$\hat{E_0} = \frac{\langle G, g\rangle_w}{\langle G, G\rangle_w}. \tag{7.3}$$

The loss-function, $\mathcal{L}$, given the correlation, is given by the subtraction of ρ^2,

$$\mathcal{L} = 1 - \rho^2(z,\gamma). \tag{7.4}$$

Once a fitted $\gamma_0, \hat{E_0}$ are chosen, the optimal z for each frame is given by a non-linear bounded least squares which are acquired using the L-BFGS-B method with a Huber Loss ($\delta = 1/2$).

When computed for each frame in the stack, one has F optimally chosen γ_0, E_0. To obtain the optimal parameter, a fit to distributions, inferred by [2], is used. These fits include a Gamma distribution for E_0 and a LogNorm distribution for γ. Together, these form nuisance estimation for E_0, γ_0.

7.6 2D Chain-link

Given an estimation of the nuisance parameters, a grid-search of the optimal z is simple. The physically best z position is a mean-squared error loss minima of the model subtracted with the data. However, due to the periodic nature of the data, there are up to 3 candidates for the optimal z position based on $\lambda/4$ having a high degree of similarity. Thus, to resolve this, a

z-linking method was implemented to find the most likely trace based on diffusion probabilities.

To link consequent positions in space, a dynamic programming algorithm is used. The algorithm is known as the Viterbi method and is described in Alg. 5.

First, all minima for every frame must be collected. These are the candidate positions that trajectories are constructed from. The losses of these positions are also stored to ensure no spurious candidates are chosen.

For each position, the forward pass compares all chains moving in time. The loss is accumulated by a weighted Euclidean transition cost,

$$d_i(i,j) = \sqrt{w_x^2(x_{t-1}^{(}i) - x_t^{(}j))^2 + w_y^2(y_{t-1}^{(}i) - y_t^{(}j))^2}, \tag{7.5}$$

where x denote z motion, and y denotes the loss difference. This matrix compares all weighted transitions. The optimal transition $\pi_t(j)$ is then chosen to minimize this transition, and any prior transitions.

For each following forward pass, the cost is accumulated with a memory of 2 frames. The memory is chosen to ensure that an incorrect transition in a given frame is not penalized later on.

Thus linking method is thus a search algorithm for the shortest math based on the weighted Euclidean method. A short toy-model is shown in Fig. 7.6 to visualize the procedure.

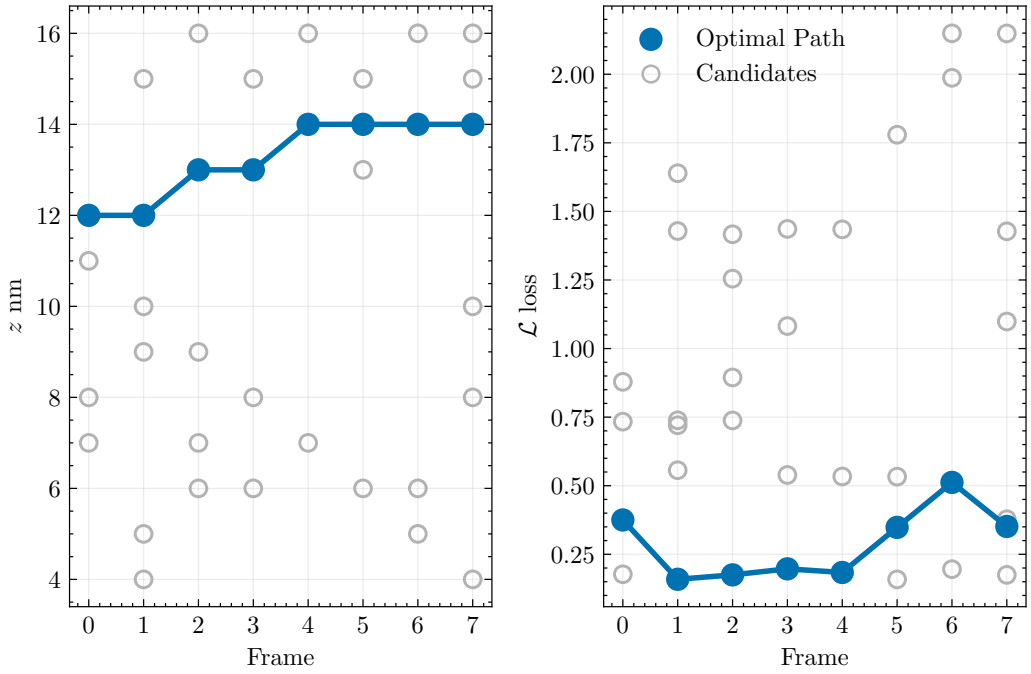


Figure 7.6: Viterbi Search algorithm toy example. Optimal path is chosen based on z similarity and lowest loss.

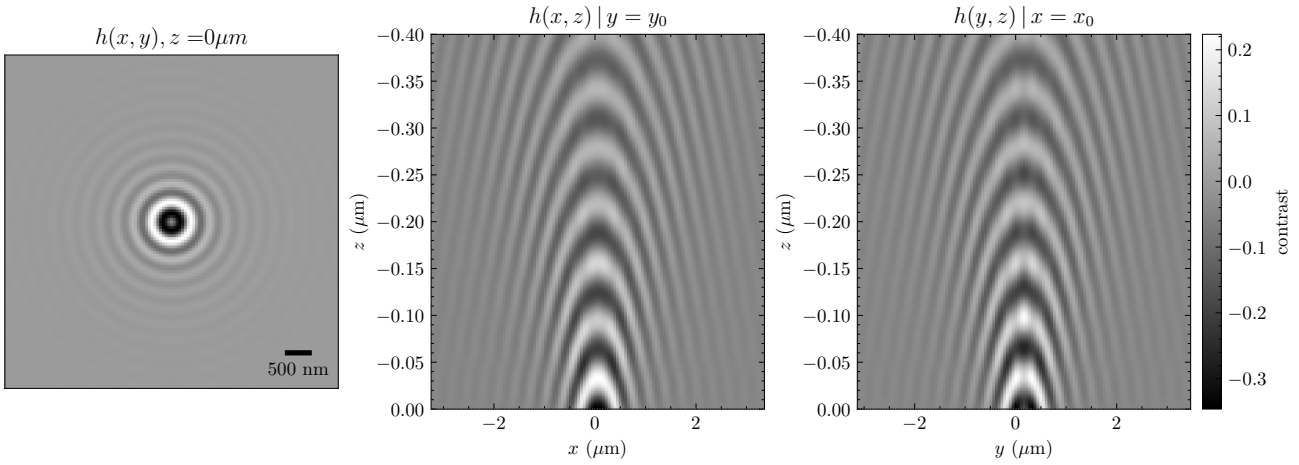


Figure 7.7: Simulated iPSF. Left. Projection of in-focus plane. Middle. x, z projection. Right. y, z projection. Focal scan is over 400 nm. pixel size is 55 nm/pixel. $\lambda = 405$ nm. Assumed particle radius $a = 50$ nm.

7.7 Simulated iPSF localization

To validate the proposed method, one performs simulated axial-estimation. Simulated iSCAT point spread functions were generated using the PySCAT library for $\lambda = 405$ nm. The iPSF was generated for the focal range $z_f \in [-1000, 0]$. The assumed particle is spherical with radius $a = 50$ nm. The iPSF is shown in Fig. 7.7

As mentioned, the localization pipeline utilizes a weighting scheme to account for objective aberration and misalignments to non-modeled regions of the data, such as the central fringe. This is accounted for by using a binary weighting array, $\mathbf{w}$. In the case of these idealized simulations, we will show the effect of changing the weighting scheme as we proceed. This demonstrates a loss of absolute precision, however, maintains the relative precision, which is that of interest in measurements of axial trajectory.

7.8 3-parameter Estimation

As a reminder, the model for the iPSF is a multivariate function iPSF : $(z, E_0, \gamma) \to \mathbb{R}^2$. It has been shown that the statistical model of iPSF takes;

$$\begin{aligned} \hat{E_0} &\sim \text{Gamma}(\mu_{E_0}, \sigma_{E_0}) \\ \hat{\gamma} &\sim \text{logNorm}(\mu_\gamma, \sigma_\gamma) \end{aligned} \tag{7.6}$$

The so-called 3-parameter estimation is a sampling of nuisance parameters in the model. The nuisance parameters here are $\hat{E_0}, \hat{\gamma}$, where $\hat{E_0}$ is an amplitude factor, and γ is an exponential decay factor used to account for spatial decay in signal. The amplitude factor can be analytically solved for using a grid-search sweep along z. The γ factor is fitted to using the L-BFGS-B

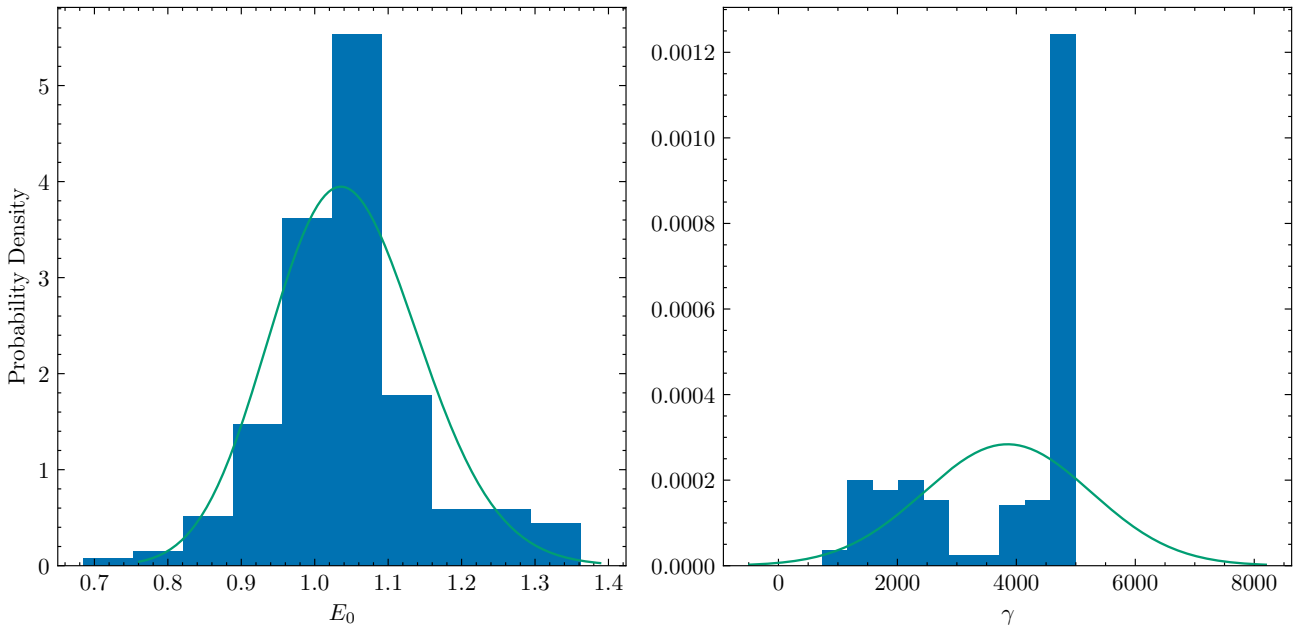


Figure 7.8: Nuisance Parameter estimation for iPSF fits. Left. Amplitude factor. right. exponential decay factor. The exponential decay is not of utmost importance as it often is very small. it is kept for stability purposes.

minimization algorithm. This nuisance parameter fit is shown in the histograms of these two parameters shown in Fig. 7.8.

7.8.1 Grid-Search Estimation

Once the nuisance parameters are chosen, a grid-search with a dynamic linking is performed. Each frame is assumed to have a maximal diffusion time. This is in place to optimize for noise and have coherent, physical traces without outliers. The grid search takes the difference between the data and a scan over the model with the nuisance parameters in place. The best loss is chosen. In the case of linking, the minimal loss over the path is chosen. The full algorithm is discussed elsewhere. the residuals of this using no fitting are shown in Fig. 7.9.

7.9 Conclusions

In conclusion, the pseudocode for the method was outlined explicitly. The methodology was tested on simulation data with a high degree of accuracy. The method shows an accuracy of $\pm 1\,\mathrm{nm}$ with a forward bias of $1\,\mathrm{nm}$.

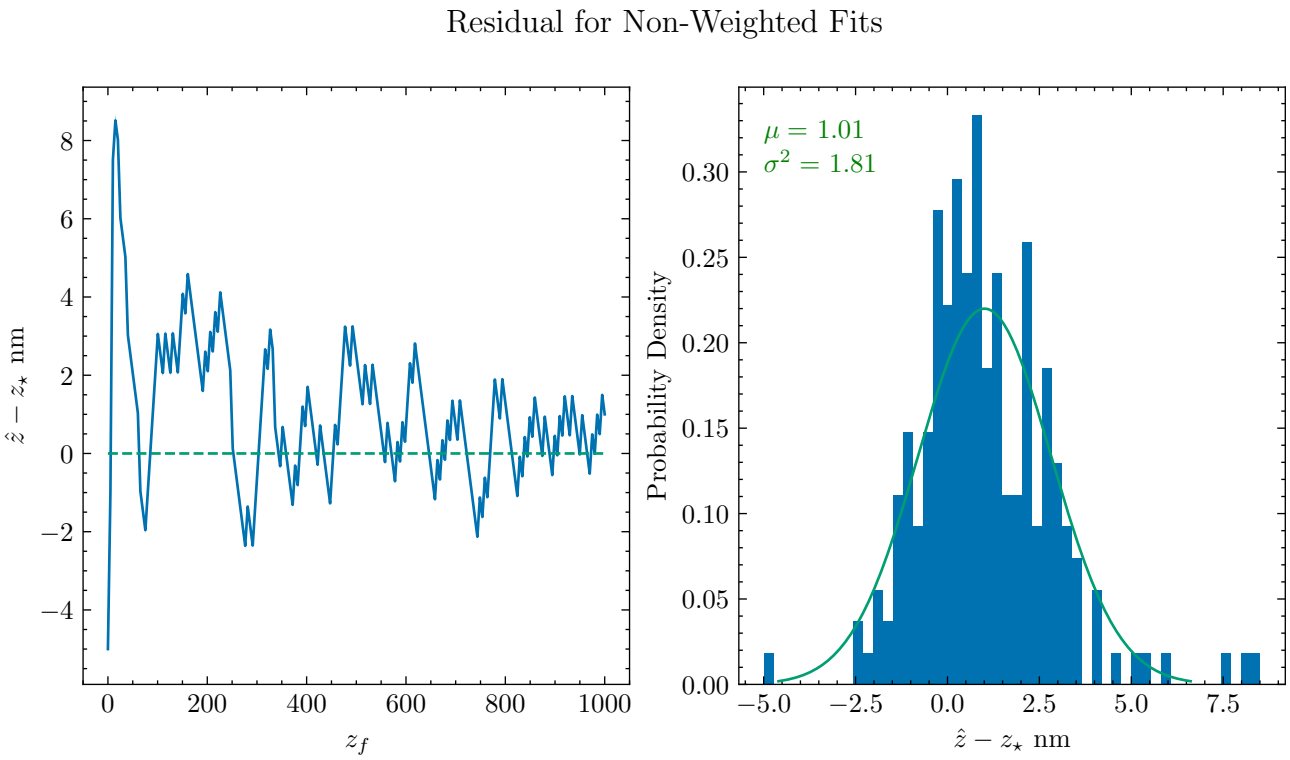


Figure 7.9: Fit residuals using no weighting scheme. reporting an absolute bias of 1 nm with variance of 2 nanometers.

Chapter 8

Experimental Results

Using iSCAT imaging technique and algorithms developed in this thesis, three different investigations were conducted.

List of Samples

1. Polystyrene Beads - 0.1 µm
   (a) Phosphate Buffer Saline 0.1%
   (b) Phosphate Buffer Saline 0.01%
   (c) Phosphate Buffer Saline 0.001%
   (d) Phosphate Buffer Saline 0.00001% - 1 height, 10 videos, 5000 frames each - 61 trajectories
2. Lipid Nanoparticle - Liposomal
3. Lipid Nanoparticle - Onpattro - 25mM NaOAc
4. Lipid Nanoparticle - Onpattro - 300mM NaOAc

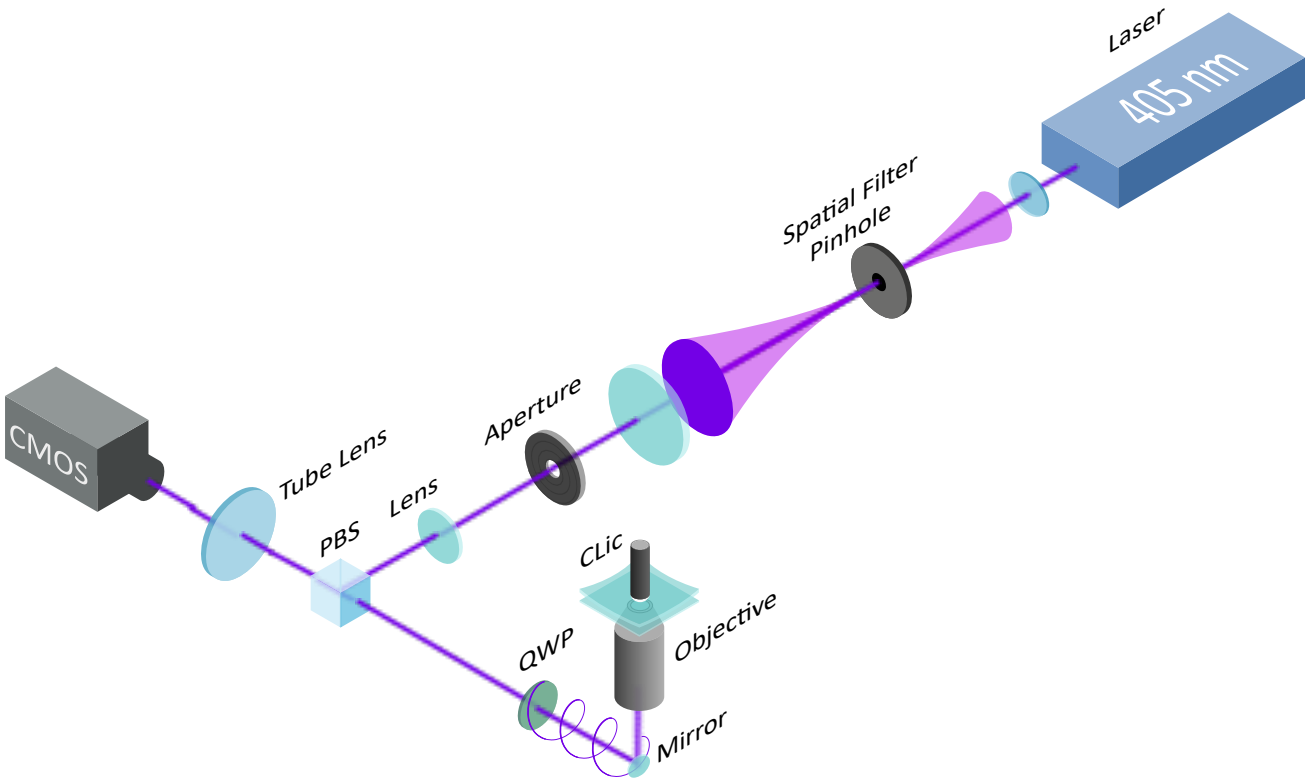


Figure 8.1: iSCAT Schematic. A 405 nm laser is sent through a spatial filter, passes through an aperture. The beam is sent through a focusing lens which is reflected off a polarizing beam splitter, polarized through a QWP focused on the back plane of the objective. The scattered light is then polarized again, passing through the polarizing beam splitter and is focused on the CMOS camera. The final image is collected via the CMOS and saved in binary format.

8.1 Methods

The following section outlines the tools used to collect data for this thesis.

8.1.1 Aparatus schematic

The schematic of the system is Here we re-draw Fig 5.1 to remind the reader of the system. The optical system used for imaging is an iSCAT microscope.

The optical system consisted of a 405 nm excitation laser. The output of the laser was 240 mW pre-spatial filter, with an output of 120 mW post-spatial filter. The objective used was the 100× oil imaging objective (Nikon, Apochromat TIRF 100× oil immersion objective N.A. 1.49, W.D. 0.12 mm, F.O.V 22 mm). The camera used was the (PRODUCT ID) CMOS camera. Operating software was custom written by (Author) in LabView.

8.1.2 Cover-Glass

The flow-cell device consisted of two glass substrates (Ted Pella, product no.260372, 145 ± 15 µm thick, 25 × 25 mm cover glass), separated by a 30 um thick double-sided adhesive spacer (Nitto

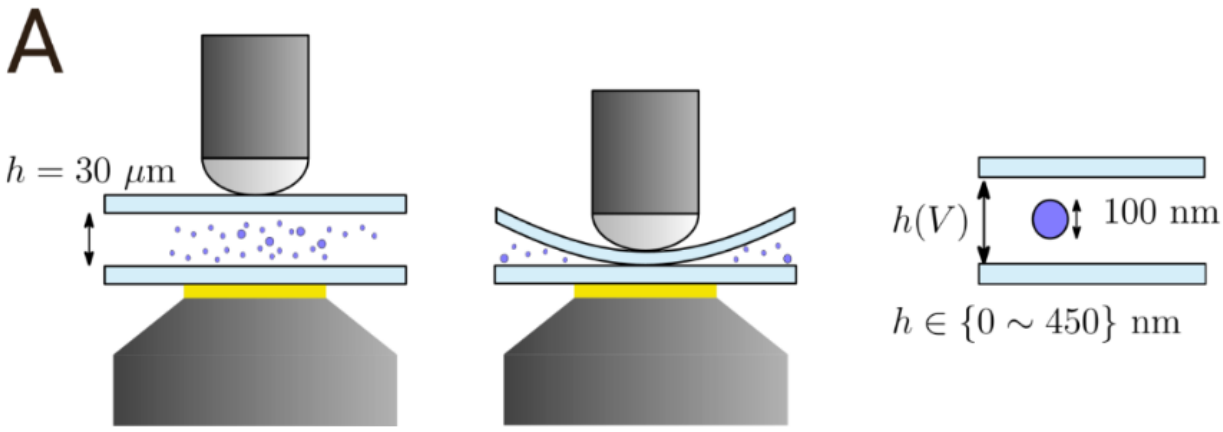


Figure 8.2: Figure Caption.

Denko, product no. 5603).

8.1.3 Convex-lens induced confinement

The CLiC system was used to achieve nanometer confinement while imaging. This method is described in detail in the seminal publication [29]. A convex pusher lens is used together with a piezo linear actuator to displace the top coverslip of the flow-cell in a parabolic displacement. The piezo actuator is manually controlled during imaging and set to a specific, controlled height. The height is determined using the inference fringes of Newton's rings. The full method outline is visualized in Fig. 8.2

8.1.4 Sample Details

The samples used in this thesis are listed below. Carboxylate modified polystyrene beads were purchased from Thermofisher (FluoSpheres™carboxylate-modified microspheres, Catalog No. F8799, are 0.1 µm polystyrene beads). Sample consists of 0.1 µm polystyrene beads with 500 µm fluorescent tag attached. The tag is attached as that is what the lab had and has used in prior measurements. The beads were able to be measured in de-ionized water (MilliQ), as well as a concentration series of phosphate-buffered saline.

Lipid-nanoparticles used in this thesis were formulated by the Pieter Cullis lab. The liposomal formulation was used as liposomal (20% norMC3). The spherical, oil-core LNP, is the Onpattro (50% norMC3) LNP formulation made with 25 mM NaOAc. The bleb-like LNP is the Onpattro bleb (50% norMC3) LNP formulation made with 300 mM NaOAc. (Cryo (?)).

8.1.5 Analysis Method

The analysis method is described in detail in prior chapters. The visual pipeline of the method includes the main inputs and outputs. The raw image is converted into a trajectory, which

includes sizing information from the MSD. This visual graphic is shown in Fig. 8.3.

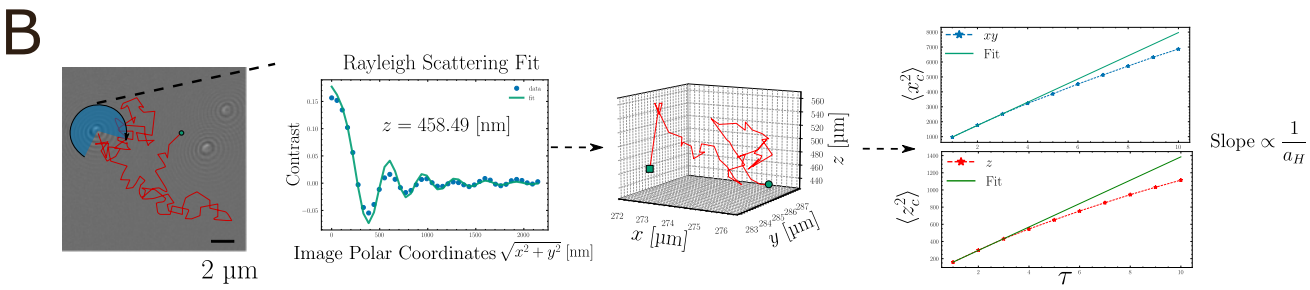


Figure 8.3: method overview.

8.2 Results

A series of experiments were conducted to determine the particle size and to validate mathematical framework presented.

To investigate effects of varied electrostatic interactions, the electrostatic screening length was varied using three concentrations of phosphate buffered saline solution. The nanoparticle used was 100 nm carboxylate-modified polystyrene beads.

To investigate the diffusivity profile of transient motion, the viscosity of the media was increased to obtain high-resolution axial trajectories. A solution of 75% glycerol with 25% 1x PBS was used to measure slow-diffusion of 100 nm carboxylate-modified polystyrene beads at a framerate of 1000 fps.

To investigate the effects of confinement on lipid nanoparticles, three formulations of LNP were imaged using a two-step passivation technique. The three-dimensional position was measured to investigate axial movement in complex media.

8.2.1 Polystyrene PBS series

The effect of ionic concentration on 100 nm polystyrene beads were performed. The beads were suspended in a series of phosphate-buffered saline (PBS) solutions with varying ionic concentrations. Ten videos were obtained according to the method described in 8.1. The resulting videos allowed to measure the three-dimensional particle position in a confined height of 220 nm. The positional probability denisty of particle motion is in Fig. 8.4.

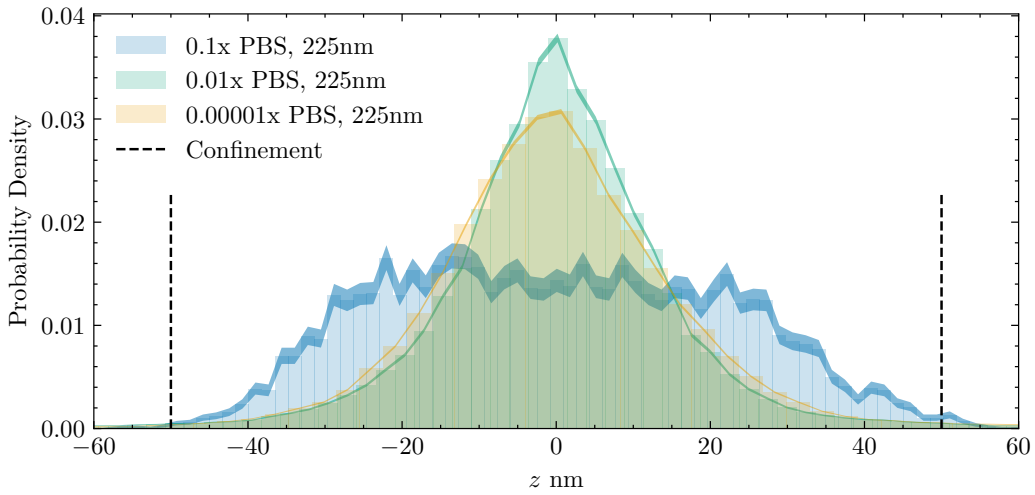


Figure 8.4: The positional histogram of particle position is plotted for three concentrations of PBS. Violet has a concentration of 0.1% PBS, green has a concentration of 0.01% PBS, and blue has a concentration of 0.00001% PBS. The horizontal axis is the axial (z-axis) position of the particle. The vertical axis is the frequency of position measurement.

Trace blue and green have the same distribution. The difference between peaks is normalization

artifact. Trace violet does not following the same distribution, but rather is flat. The diffusion in blue and green is screened, where the particle does not explore the full height. In violet the particle does explore the well more. As the concentration of PBS increases, the distribution becomes more uniform, while in the higher concentrations, the particle is focused between the two plates.

This experiment agrees with the simulations done as shown in Fig. 6.2.

The videos are large ($\sim 30\,\mathrm{Gb}$ per video) and are contained in a zip file upon request.

For each measurement, 100 nm diameter beads were confined into a 220 nm slit verified using inference fringes.

For each vide

The sizes recovered from a transverse diffusion MSD method and sizes recovered from the axial traveled distance were recovered and compared.

8.2.2 Polystyrene Transient Diffusion

For each plot, the theoretical curve predicted by the linear-superposition theory [1] is presented. Measured single-molecule data is presented overtop this. Axial positions are binned into tracks and diffusivity is measured on these tracks. This is plotted for single-molecule data and ensemble averages. These plots are shown in Fig. 8.5

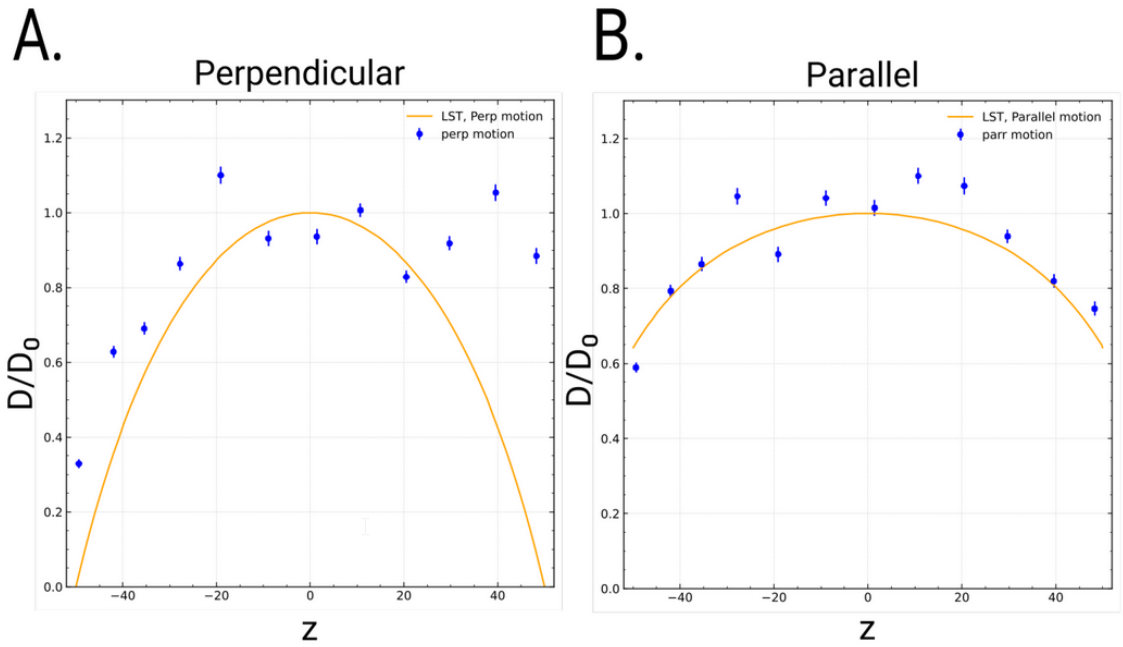


Figure 8.5: Figures Description

8.2.3 Lipid nanoparticle Diffusion

LNP diffusivity is evaluated for various LNP formulations... This is a WIP but the data has been taken already

8.3 Discussion of Results

The discussion starts here

Chapter 9

Conclusion

9.1 Conclusion

The introduction introduces the topics of Brownian motion in a confined space, because this is an active field of research with direct application to

This thesis presents the current state of research in the features of nanoparticle diffusion in confinement and single-particle tracking using interferometric light scattering microscopy. The theory of Brownian motion is well described by the Langevin framework.

The theory of thermal motion is reviewed based on existing research publications and provides

9.2 Future outlook

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Appendix A

Surface Charge Estimator Derivation

This is a section for the derivation of the surface charge estimator used in the axial diffusion estimation.

A.1 $\hat{B}$ MLE

The Force used in the modelling is given by;

$$V'(z) = B\frac{k_bT}{\lambda_D} \cdot \exp(-H/\lambda_D) \cdot (\exp(-z_n/\lambda_D) - \exp(z_n/\lambda_D)) \tag{A.1}$$

The dynamics are governed by the equation, written in the Ito convention,

$$\Delta z = D'(z)\Delta t - \beta D(z)V'(z)\Delta t + \sqrt{2D_z}W_t. \tag{A.2}$$

Re-writing in terms of drift and diffusion coefficients;

$$\Delta z = \left[D_z'(z) - \beta BD(z)\phi(z)\right]\Delta t + \sqrt{2D_z}W_t. \tag{A.3}$$

where $\phi(z) = \frac{k_bT}{\lambda_D}\exp(-H/\lambda_D) \cdot (\exp(-z_n/\lambda_D) - \exp(z_n/\lambda_D))$. This is a Markov-process that has a propagator probability from the Onsager-Machlup path integral representation,

$$P(z_{n+1}|z_n) = \frac{1}{4D_z\Delta t}\exp\left[-\frac{\left(\Delta z_n - D_z'(z) + \beta BD(z)\phi(z)\right)^2}{4D_z\Delta t}\right]. \tag{A.4}$$

To obtain an estimator for B, we can take the log-likelihood and differentiate, maximizing it.

$$
\begin{aligned}
l(B) &= \sum_{n=0}^{\infty} \left[-\frac{\left(\Delta z_n - D_z'(z) + \beta B D(z)\phi(z)\right)^2}{4D_z\Delta t} \right] + \text{const} \\
&= -\frac{1}{2}\sum_{n=0}^{\infty} \frac{1}{2D_z\Delta t} \left(\Delta z_n - D_z'(z) + \beta B D(z)\phi(z)\right)^2 + \text{const.}
\end{aligned}
$$

Define $r_n = \Delta z_n - D_z'(z)$ and $w_n = \frac{1}{2D_z}$. Find the maximum of the log-likelihood by taking the derivative of l with respect to B.

$$
l'(B) = -\sum_{n=0}^{\infty} \frac{w_n}{\Delta t} \left(r_n + \beta B D(z)\phi(z)\right) \beta D(z)\phi(z)
$$

We can set this to 0 and solve for $\hat{B}$.

$$
\hat{B} = \frac{\sum_n \frac{\phi(z)}{2D_z} \left[\Delta z - D'(z)\Delta t\right]}{\sum_n \frac{\phi^2(z)\Delta t}{2D_z}} \tag{A.5}
$$

A.2 $\mathbb{V}[\hat{B}]$

The variance of this can be found using the fisher information matrix;

$$
\begin{aligned}
\mathcal{I}(B) &= -l''(B) \\
&= \sum_{n=0}^{\infty} \frac{\Delta t \phi^2(z)}{2D(z)}
\end{aligned}
$$

In the univariate case, the variance is proportional to the inverse of the fisher information, or the denominator of the MLE.

$$
\mathbb{V}[\hat{B}] = \sum_{n=0}^{\infty} \frac{\phi^2(z)\Delta t}{2D(z)} \tag{A.6}
$$

This is an appication of the delta-method as

$$
\begin{aligned}
W_t &\sim \mathcal{N}(0,1) \\
B &= g(W_t).
\end{aligned} \tag{A.7}
$$

Appendix B

Axial Estimation Algorithm Pseudocode

This section of the appendix lists all detailed algorithms and computational methods presented in the axial-estimation section of this project.

The following algorithms are presented:

Function	Desc.
Raw2iScat:	Camera image to iSCAT contrast stack.
TransverseLocalization:	(x, y) localization function with sub-pixel resolution.
RadialIntegral:	Radial integral function.
Nuisance-parameter estimation:	Search algorithm with a grid-scan and raw-python implemented least-squares for Model.
Axial Chain-Link:	Linking algorithm for axial trajectories. Analagous to Munkre's algorithm in 2D tracking.

Table B.1: List of functions presented in appendix. Function names and descriptions are listed.

Each algorithm is presented in language-general mathematical pseudocode. Verbal explanations are described in chapter 7. The appendix serves as a more-detailed explanation so that the interested reader may implement the software themselves and have exact computational details.

A Python implementation, written by the author and open-source contributions, can be produced upon request.

Algorithm 1 Raw image stack → iSCAT contrast stack

Require: $I_{\text{raw}} \in \mathbb{R}^{F\times H\times W}$ ▷ F frames of $H \times W$ pixels
Ensure: $I_{\text{iSCAT}} \in \mathbb{R}^{F\times H\times W}$

Parameters

$R_1 \leftarrow \max(H,W)/100, \quad R_2 \leftarrow \max(H,W)/4$ ▷ band-pass radii in k-space [px]
$\Delta f \leftarrow 25, \quad \Delta L \leftarrow 15$ ▷ temporal gap; background half-window, empirical for data conditions.
$N \leftarrow$ border width ▷ pixels discarded at the edges

Pre-computation (frame independent)

$(k_x, k_y) \leftarrow \textsc{Meshgrid}\big(\big[-\frac{H}{2}, \frac{H}{2}\big), \big[-\frac{W}{2}, \frac{W}{2}\big)\big)$ ▷ centred spatial frequencies
$h \leftarrow \mathbf{1}\big[\, R_1^2 < k_x^2 + k_y^2 < R_2^2 \,\big]$ ▷ annular band-pass filter
$M_{pq} \leftarrow \mathbf{1}[\, N \le p < H - N \ \wedge \ N \le q < W - N \,]$ ▷ edge mask

function $\textsc{AmpNorm}(I, M)$ ▷ row-wise amplitude normalisation of one frame. Row-wise means to avoid readout noise from the camera. Mean over rows to remove line artifacts. (From Vahid's Method)
    $I^{(1)} \leftarrow I \odot M$ ▷ $\odot$: element-wise (Hadamard) product
    $I^{(2)} \leftarrow I^{(1)} / \langle I^{(1)} \rangle$ ▷ $\langle\cdot\rangle$: mean over all pixels
    for each row $p \in \{0, \dots, H-1\}$ **do**
        $a_p \leftarrow \langle M_{p,:} \rangle / \langle I^{(2)}_{p,:} \rangle$ ▷ row means
        $I^{(3)}_{p,:} \leftarrow I^{(2)}_{p,:} / a_p$
    end for
    return $I^{(3)}$
end function

Main loop

for $i = 0$ **to** $F - 1$ **do**
    $I_0 \leftarrow \textsc{AmpNorm}(I_{\text{raw}}[i],\, M)$ ▷ signal frame

    $\mathcal{B}_i \leftarrow \big[\, i - \Delta f - \Delta L,\ i - \Delta f \,\big] \ \cup \ \big[\, i + \Delta f,\ i + \Delta f + \Delta L \,\big]$ ▷ background frame indices
    $\mathcal{B}_i \leftarrow \mathcal{B}_i \cap [\, 0, F \,)$ ▷ clip at the ends of the stack
    $\bar{B} \leftarrow \dfrac{1}{|\mathcal{B}_i|} \displaystyle\sum_{j \in \mathcal{B}_i} \textsc{AmpNorm}(I_{\text{raw}}[j],\, M)$ ▷ mean background
    assert $\bar{B}$ contains no NaN

    $I_m \leftarrow \big(I_0 \oslash \bar{B}\big) \odot M$ ▷ $\oslash$: element-wise division
    $I_{\text{iSCAT}}[i] \leftarrow \mathcal{F}^{-1}\{h \odot \mathcal{F}\{I_m\}\}$ ▷ band-pass filtering
end for

return I_{iSCAT}

Algorithm 2 Transverse Localization algorithm

Inputs
I_{iSCAT} ▷ from Alg. 1
$I_{\text{RVT}} \leftarrow \text{RVT}[I_{\text{iSCAT}}]$

Pre-computation (frame independent)
$t \leftarrow 0.9 \cdot \max I_{\text{iSCAT}}$ ▷ threshold
$n \leftarrow 30$ ▷ number of particles per frame to assume, empirical
$x_{\text{ROI}}, y_{\text{ROI}} \leftarrow 5$ ▷ ROI cropping for radial symmetry algorithm
Fitting Routine
for $i = 0$ **to** $F - 1$ **do**
$I_{\text{dil}} \leftarrow \textsc{ImDialate}(I_{\text{RVT}}[i])$
$\{x, y\} \leftarrow \textsc{Where}\,[I_{\text{dil}} == I_{\text{RVT}} \wedge I_{\text{RVT}} > t]$ ▷ Candidate Positions
$\{x_1, y_1\}|_n = \textsc{GaussianFit}(x, y)$ ▷ Non-linear least squares fitting, take best n fits
$\{x_\star, y_\star\} = \textsc{RadialSymmetry}(x_1, y_1, x_{\text{ROI}}, y_{\text{ROI}})$ ▷ from [27]
end for
2D Trace Linking
Traces $\leftarrow \textsc{Munkres}[x_\star, y_\star]$ ▷ Minimization of distance cost-matrix
ROI $= I_{\text{iSCAT}}[Traces]$ ▷ Cropping of image around every frame for each trace.

Algorithm 3 Discrete radial profile (angular average over a wedge)

Require: $I \in \mathbb{R}^{F\times H\times W}$ ▷ stack of F image ROIs
Require: $\mathbf{c} = (c_p, c_q)$ ▷ ROI centre (may be sub-pixel)
Require: θ_0, $\Delta\theta$ ▷ start angle and angular width of the wedge
Require: R ▷ outer radial bound, in pixels
Ensure: $g \in \mathbb{R}^{F\times R}$, where

$$g_t(\rho) \;\simeq\; \frac{1}{\Delta\theta}\int_{\theta_0}^{\theta_0+\Delta\theta} I_t(\rho,\theta)\,\mathrm{d}\theta$$

is the angular mean of frame t over the wedge, as a function of radius.

Pre-computation (frame independent)

$\rho_{pq} \leftarrow \big\|(p,q)-\mathbf{c}\big\|_2, \quad \theta_{pq} \leftarrow \operatorname{arctan2}\big(p-c_p,\; q-c_q\big) \bmod 2\pi$ ▷ polar coordinates of each pixel

for $r = 0$ **to** $R-1$ **do** ▷ annular-wedge bins
  $\mathcal{A}_r \leftarrow \big\{\,(p,q)\;:\; r \le \rho_{pq} < r+1 \;\wedge\; \big(\theta_{pq}-\theta_0\big) \bmod 2\pi \;<\; \Delta\theta\,\big\}$
  $N_r \leftarrow |\mathcal{A}_r|$ ▷ $N_r \approx (r+\frac{1}{2})\,\Delta\theta$
end for

Main loop

for $t = 0$ **to** $F-1$ **do**
  for $r = 0$ **to** $R-1$ **do**
    if $N_r = 0$ **then**
      $g_t(r) \leftarrow \textsc{NaN}$ ▷ empty bin: undefined, not zero
    else
      $g_t(r) \leftarrow \dfrac{1}{N_r}\displaystyle\sum_{(p,q)\in\mathcal{A}_r} I_t[p,q]$ ▷ angular average at radius r
    end if
  end for
end for

return g

Algorithm 4 Nuisance-parameter estimation by profiled grid search + local refinement

Require: $g \in \mathbb{R}^M$ ▷ radial profile, sampled at radii r_i, $i = 1, \dots, M$
Require: $r \in \mathbb{R}^M$, $w \in \mathbb{R}^M_{\geq 0}$ ▷ radii [nm] and fit weights
Require: $k = 2\pi n_0/\lambda$ ▷ optical wave-vector in the medium
Require: $[z_-, z_+]$, $[E_-, E_+]$, $[\gamma_-, \gamma_+]$ ▷ parameter bounds
Require: $\Delta_\lambda = 32$

iPSF Model

$G_i(z, \gamma) = \frac{q(r_i,z)}{\varrho(r_i,z)^2} \cdot \cos\big[\, k\,(z + \varrho(r_i, z))\,\big]\, e^{-r_i/\gamma}, \quad \varrho(r, z) = \sqrt{r^2 + z^2}$

$\chi^2(E_0, z, \gamma) = \big\langle\, g - E_0 G,\ g - E_0 G \,\big\rangle_w, \quad \langle a, b\rangle_w \equiv \sum_{i=1}^M w_i\, a_i\, b_i$

Pre-computation

$\delta_z \leftarrow 2\pi/(k\,\Delta_\lambda)$ ▷ $= (\lambda/n_0)/\Delta_\lambda$: fringe period / Δ_λ

$\mathcal{Z} \leftarrow \{\, z_- + m\,\delta_z \ :\ m = 0, \dots, N_z - 1 \,\}, \quad N_z = \lceil (z_+ - z_-)/\delta_z \rceil$

$\Gamma \leftarrow \{\, \gamma_1, \dots, \gamma_{N_\gamma} \,\} \subset [\gamma_-, \gamma_+]$ ▷ decay-length grid

$\langle g, g\rangle_w \leftarrow \sum_i w_i\, g_i^2$

Profiled grid search

for $\gamma \in \Gamma$ **do**
  for $z \in \mathcal{Z}$ **do**
    $\varrho_i \leftarrow \sqrt{r_i^2 + z^2} \quad \forall\, i$
    $G_i \leftarrow \dfrac{q(r_i, z)}{\varrho_i^2}\, \cos\big[k(z + \varrho_i)\big]\, e^{-r_i/\gamma} \quad \forall\, i$
    $\langle G, G\rangle_w \leftarrow \sum_i w_i\, G_i^2, \quad \langle G, g\rangle_w \leftarrow \sum_i w_i\, G_i\, g_i$
    $\rho(z, \gamma) \leftarrow \dfrac{\langle G, g\rangle_w}{\sqrt{\langle G, G\rangle_w\, \langle g, g\rangle_w \ +\ \varepsilon}}$ ▷ weighted correlation coefficient
    $\hat{E}_0(z, \gamma) \leftarrow \textsc{Clip}\left[\, \dfrac{\langle G, g\rangle_w}{\langle G, G\rangle_w},\ E_-,\ E_+ \right]$ ▷ profiled amplitude
    $\mathcal{L}(z, \gamma) \leftarrow 1 - \rho^2(z, \gamma)$ ▷ $\chi^2_{\min}/\langle g, g\rangle_w$
  end for
end for

Refinement

$(z^\star, \gamma^\star) \leftarrow \underset{(z,\gamma)\,\in\,\mathcal{Z}\times\Gamma}{\arg\min}\ \mathcal{L}(z, \gamma)$ ▷ coarse global minimum

$E_0^\star \leftarrow \hat{E}_0(z^\star, \gamma^\star)$

$(\hat{E}_0, \hat{z}, \hat{\gamma}) \leftarrow$ L-BFGS-B$\big[\, \sum_i \mathcal{H}_\delta\big(g_i - E_0 G_i(z, \gamma)\big);$ init $(E_0^\star, z^\star, \gamma^\star);$ bounds $[E_\pm], [z_\pm], [\gamma_\pm]\,\big]$
▷ Huber loss $\mathcal{H}_\delta$

return $(\hat{E}_0,\ \hat{z},\ \hat{\gamma})$

Algorithm 5 Dynamic program for the optimal 2D chain of candidate points, Viterbi Algorithm

Require: Candidate sets $\mathcal{C}_t = \{\, \mathbf{p}_t^{(i)} = \big(x_t^{(i)},\, y_t^{(i)}\big) \,\}_{i=1}^{n_t},\ t = 0, \dots, F-1$
where $x_t^{(i)} \equiv \texttt{min_idx}[t][i]$ (position) and $y_t^{(i)} \equiv \texttt{min_vals}[t][i]$ (value)
Require: Weights $\mathbf{W} = \mathrm{diag}(w_x, w_y)$, default $\mathbf{W} = \mathbf{I}$
Require: Maximum step $s_{\max} \in (0, \infty]$ ▷ $s_{\max} = \infty \Rightarrow$ unconstrained
Ensure: Chain $\mathbf{k} = (k_0, \dots, k_{F-1})$, $k_t \in \{1, \dots, n_t\}$, minimizing

$$J(\mathbf{k}) \;=\; \sum_{t=1}^{F-1} \big\|\mathbf{W}\big(\mathbf{p}_{t-1}^{(k_{t-1})} - \mathbf{p}_t^{(k_t)}\big)\big\|_2 \qquad \text{s.t.} \quad \big\|\mathbf{W}\big(\mathbf{p}_{t-1}^{(k_{t-1})} - \mathbf{p}_t^{(k_t)}\big)\big\|_2 \le s_{\max}$$

together with the corresponding x- and y-chains.

Forward pass ▷ $C_t(j)$: cost of the best chain ending at $\mathbf{p}_t^{(j)}$

$C_0(i) \leftarrow 0 \quad \forall\, i \in \{1, \dots, n_0\}$
for $t = 1$ **to** $F-1$ **do**
▷ Weighted Euclidean transition cost, $n_{t-1} \times n_t$
  $d_t(i,j) \leftarrow \sqrt{w_x^2\big(x_{t-1}^{(i)} - x_t^{(j)}\big)^2 + w_y^2\big(y_{t-1}^{(i)} - y_t^{(j)}\big)^2} \quad \forall\,(i,j)$
  $d_t(i,j) \leftarrow \infty$ **if** $d_t(i,j) > s_{\max}$ ▷ forbid over-long steps

▷ Bellman recursion
  $\pi_t(j) \leftarrow \arg\min_i \big[\, C_{t-1}(i) + d_t(i,j) \,\big] \quad \forall\, j \in \{1, \dots, n_t\}$
  $C_t(j) \leftarrow C_{t-1}\big(\pi_t(j)\big) + d_t\big(\pi_t(j),\, j\big)$

  if $C_t(j) = \infty\ \forall\, j$ **then**
    return $\varnothing$ ▷ no admissible chain survives frame t
  end if
end for

Backtracking
$k_{F-1} \leftarrow \arg\min_j C_{F-1}(j)$ ▷ cheapest terminal node
for $t = F-1$ **down to** 1 **do**
  $k_{t-1} \leftarrow \pi_t(k_t)$
end for

return $\big\{k_t\big\}, \quad \big\{x_t^{(k_t)}\big\}, \quad \big\{y_t^{(k_t)}\big\} \quad$ for $t = 0, \dots, F-1$